STATISTICS 475: Applied Discrete Data Analysis

Lecture 22: Poisson regression models for three or more variables

(B&L Section 4.2.5)

1 Problem to be solved

- We have seen how to write a Poisson loglinear regression model to model and measure associations between two categorical variables
- Now consider modeling associations among 3 or more variables
- Here is where Poisson loglinear models begin to show their flexibility
  - They also begin to show their complexity.
    - Complication is in the *huge* array of models that can be represented using these techniques
    - Nothing new about estimation and inference
- **TOM: DRAW LOTS OF PICTURES!!!**

2 Loglinear model for 3 variables

Example: Political Ideology (PolIdeolNominal.R, PolIdeolData.csv)

(Straight out of the book) Agresti (2007) presents a summary from the 1991 U.S. General Social Survey[1] cross-classifying people according to

[1]Agresti, A. (2007). *An Introduction to Categorical Data Analysis, 2nd ed.* New York: Wiley. Note that the source of these data is a large-scale survey with a complex probability-based sampling design. Failing to account for the survey design in the analysis could lead to biased parameter estimates and misrepresentation of the true relationships among the variables. Proper techniques for analyzing data from this type of survey are presented in Section 6.3 and should be applied here. The analyses we present here are used solely as an illustration of the present analysis techniques and as a comparison to previously published work.

- political ideology, `ideol`, a 5-level variable with levels 1: "Very Liberal" (`"VL"`), 2: "Slightly Liberal" (`"SL"`), 3: "Moderate" (`"M"`), 4: "Slightly Conservative" (`"SC"`), and 5: "Very Conservative" (`"VC"`)
- political party, `party`, a 2-level variable with levels "Republican" (`"R"`) and "Democrat" (`"D"`)
- gender, `gender`, a 2-level variable with levels "Female" (`"F"`) and "Male" (`"M"`); no other genders were recorded.

There are numerous questions one could ask about data like this. We already know that Republicans in the US tend to be more conservative and the Democrats more liberal. But are women more or less conservative than men? Are they more often Republican than Democrat? And, a little more deeply, *is the known, strong relationship between party and ideology the same for both genders*?

We will fit a variety of models to these data. They will differ in their interpretation of which variables are associated and whether the associations are the same or different across levels of other variables. Notice that the ideology variable is ordinal. In the next lecture, we explore how that structure can be exploited to provide a more parsimonious description of the association. For now, we treat ideology as nominal. Below is part of the data:

```
> alldata <- read.table("C:\\Data\\PolIdeolData.csv", sep = ",",
                header = TRUE, stringsAsFactors=TRUE)
> head(alldata)
  gender party ideol count
1      F     D    VL    44
2      F     D    SL    47
3      F     D     M   118
4      F     D    SC    23
5      F     D    VC    32
6      F     R    VL    18
>
> # R will alphabetize levels of "ideol".  Don't want that.
> alldata$ideol <- factor(alldata$ideol,
                     levels=c("VL", "SL", "M", "SC", "VC"))
> # Show tables of the data
> # In xtabs, first variable is row, second is column,
> #    third and later arrange separate tables.  So code below
> #    tabulates ideology against party separately by gender
> tab.gpi <- xtabs(formula= count ~ party + ideol + gender,
                     data=alldata)
> tab.gpi
, , gender = F

     ideol
party  VL  SL   M  SC  VC
```

```
    D   44   47  118   23   32
    R   18   28   86   39   48

, , gender = M

      ideol
party   VL   SL    M   SC   VC
    D   36   34   53   18   23
    R   12   18   62   45   51
```

I have used `xtabs()` to create tables of the data in a way that shows some of the expected trends.

2.1 Questions and Notation

- When we model counts for categorical variables, the goal is to learn about associations among variables
  - "Association" means that conditional distribution of levels of one variable changes depending on levels of the other variable.
    - Causes ratios of means from two different levels of one variable to change depending on levels of other variable
    - Measured by comparing mean ratios like

$$\frac{\mu_{11}/\mu_{12}}{\mu_{21}/\mu_{22}}$$

    - This is an odds ratio
    - No association means that all odds ratios in a table are 1
- When we have more than two variables, we want to know which pairs of variables are associated and which are not
- Another question: Are associations between two variables consistent across levels of other variables, or do they change?
  - In our example, is the party-ideology relationship the same for both genders?
  - Are odds ratios involving the two parties and any two ideologies always the same across the genders?
- More generally, let's suppose we have three explanatory variables
  - X has levels $i = 1, \ldots, I$; Z has levels $j = 1, \ldots, J$; and W has levels $k = 1, \ldots, K$
- A count y_{ijk} is observed at each combination (i, j, k) of X, Z, W

- A table of these counts requires 3 dimensions (Rubik's Cube)

- We assume that the count RV $Y_{ijk} \sim Po(\mu_{ijk})$
  - The question is how the mean counts in the cells, μ_{ijk}, relate to the explanatory variables:
$$\log(\mu_{ijk}) = \beta_0 + \text{possible other stuff}$$
  - What is the "other stuff"?
- We know that each explanatory variable is replaced in a model by indicators
  - X: $I - 1$ indicators $x_1, x_2, \ldots, x_I$
  - Z: $J - 1$ indicators $z_1, z_2, \ldots, z_J$
  - W: $K - 1$ indicators $w_1, w_2, \ldots, w_K$
- R will delete the first indicator of each sequence, but it is more convenient for notation to leave them in place and assign their parameters to be 0
- We put all of these indicators into a model with parameters
  - For X we have $\beta_1^X, \beta_2^X, \ldots, \beta_I^X$, (fixing $\beta_1^X = 0$ and leaving $I - 1$ parameters to be estimated). Other variables done similarly.
  - For any particular cell in the table, (i, j, k), only the indicators $x_i, y_j,$ and z_k are 1 and the rest are 0
  - Thus, the model for any specific mean μ_{ijk} uses only
$$\log(\mu_{ijk}) = \beta_0 + \beta_i^X + \beta_j^Z + \beta_k^W,$$
    and all other parameters are 0.
    * If any of $i, j,$ or k are 1, then these parameters end up being 0 as well.
- In general, we write loglinear models in this abbreviated way, so that we don't have to write out the long, boring regression models

2.2 Model Terms and Meanings

- Recall that in the $I \times J$ tables, there are observed counts y_{ij}, row marginal counts y_{i+} and column marginal counts y_{+j}
  - Lecture 21 we treated each cell count as $Po(\mu_{ij})$ where each cell has its own mean μ_{ij},
    * From this there were row marginal means μ_{i+}, and column marginal means μ_{+j} for each row i and column j.
  - The model with main effects ("first-order" terms) of each variable,
  $$\log(\mu_{ij}) = \beta_0 + \beta_i^X + \beta_j^Z,$$
  ensured that the model-estimated marginal counts matched the table marginal counts, but did not perfectly model the individual cells
    * $\hat{\mu}_{i+} = y_{i+}$ for each row i, and $\hat{\mu}_{+j} = y_{+j}$ for each column j
    * But not necessarily true that $\hat{\mu}_{ij} = y_{ij}$
    * All odds ratios in estimated table are 1
  - Adding interaction ("second-order" terms) to the model forced the model to have as many parameters as the table and, thus, each cell is perfectly estimated
    * $\hat{\mu}_{ij} = y_{ij}$, model is "saturated" and has zero deviance
    * Odds ratios can differ from 1

- **With three variables, we can have more of the same kinds of main effects and interaction terms in the model**

  1. There is a third main effect term:
  $$\log(\mu_{ijk}) = \beta_0 + \beta_i^X + \beta_j^Z + \beta_k^W$$
  for each $i = 1, \ldots, I$; $j = 1, \ldots, J$; $k = 1, \ldots, K$, representing the indicator variables in the regression model
     - These terms only ensure that the model matches the total (marginal) counts for *each level of each variable*
       * In notation, $\hat{\mu}_{i++} = y_{i++}$ for each i, $\hat{\mu}_{+j+} = y_{+j+}$ for each j, and $\hat{\mu}_{++k} = y_{++k}$ for each k, where the double + means computing sums over all levels of the variables for the respective subscripts.
       * For a fixed level of one variable, all ORs between two rows and two columns of the resulting table are 1.
     - Referred to as MUTUAL INDEPENDENCE MODEL or FIRST-ORDER MODEL
  2. There are now 3 different two-way interactions (added into the model with main effects):
  $$\log(\mu_{ijk}) = \beta_0 + \beta_i^X + \beta_j^Z + \beta_k^W + \beta_{ij}^{XZ} + \beta_{jk}^{XW} + \beta_{jk}^{ZW}$$
  for each $i = 1, \ldots, I$; $j = 1, \ldots, J$; $k = 1, \ldots, K$ representing cross products of all pairs of indicator variables in the regression model

- These terms ensure that the model matches the total (marginal) counts for *each combination of levels of each pair of variables*
- Allows odds ratios between pairs of levels for two variables to differ from 1, but they are constant across levels of the third variable
  - β_{ij}^{XZ} terms control XZ log-ORs:

$$\log(OR_{i_1 i_2 \backslash j j_2 | k}^{XZ}) = \log\left(\frac{\mu_{i_1 j_1 k}/\mu_{i_1 j_2 k}}{\mu_{i_2 j_1 k}/\mu_{i_2 j_2 k}}\right) = \beta_{i_1 j_1}^{XZ} - \beta_{i_1 j_2}^{XZ} - \beta_{i_2 j_1}^{XZ} + \beta_{i_2 j_2}^{XZ},$$

    for any two levels i_1 and i_2 of X, two levels j_1 and j_2 of Z, and any level k of W
  - Similar for XW and ZW ORs.
- Referred to as HOMOGENEOUS ASSOCIATION MODEL or SECOND-ORDER MODEL

3. (NEW!) Three-way interactions:

$$\log(\mu_{ijk}) = \beta_0 + \beta_i^X + \beta_j^Z + \beta_k^W + \beta_{ij}^{XZ} + \beta_{jk}^{XW} + \beta_{jk}^{ZW} + \beta_{ijk}^{XZW}$$

for each $i = 1, \ldots, I$; $j = 1, \ldots, J$; $k = 1, \ldots, K$ representing cross products of indicator variables from all three variables (i.e., terms like $\beta_{243}^{XZW} x_2 z_4 w_3$)

- These terms ensure that the model is saturated so that each $\hat{\mu}_{ijk} = y_{ijk}$
  - Deviance is zero. All degrees of freedom consumed by the model
  - IJK parameters!
- Allows odds ratios between pairs of levels for two variables to change depending on the level of the third variable
  - β_{ij}^{XZ} terms control XZ ORs for the first level of W, and β_{ijk}^{XZW} terms control XZ ORs for the other levels:

$$\log(OR_{i_1 i_2, j_1 j_2, k}^{XZ}) = \log\left(\frac{\mu_{i_1 j_1 k}/\mu_{i_1 j_2 k}}{\mu_{i_2 j_1 k}/\mu_{i_2 j_2 k}}\right) = (\beta_{i_1 j_1}^{XZ} - \beta_{i_1 j_2}^{XZ} - \beta_{i_2 j_1}^{XZ} + \beta_{i_2 j_2}^{XZ})$$
$$+(\beta_{i_1 j_1 k}^{XZW} - \beta_{i_1 j_2 k}^{XZW} - \beta_{i_2 j_1 k}^{XZW} + \beta_{i_2 j_2 k}^{XZW})$$

    for any two levels i_1 and i_2 of X, two levels j_1 and j_2 of Z, and any level k of W
  - Similar for XW ORs at levels of Z and ZW ORs at levels of X.
- Referred to as HETEROGENEOUS ASSOCIATION MODEL or THIRD-ORDER MODEL (or just SATURATED MODEL)

- ALL of these properties are found just by taking differences of log-means (linear predictors) *very carefully!*
  - OR formulas are not difficult to compute if you are careful
  - **`emmeans` package does all of this for you! You just have to know how to ask**

- Degrees of Freedom for each term = number of new parameters generated by the indicators in term
  - $(I-1)$, $(J-1)$, and $(K-1)$ for main effects
  - $(I-1)(J-1)$, $(I-1)(K-1)$, and $(J-1)(K-1)$ for 2-way interaction terms.
  - $(I-1)(J-1)(K-1)$ for 3-way interaction

<u>Example</u>: Political Ideology (PolIdeolNominal.R, PolIdeolData.csv)

We will represent the three factors by their first letters: G for `gender`, P for `party`, and I for `ideol`. We already know that PI should be important, so our main goals are

1. Determine which associations among GI, GP may be important,
2. Determine whether the expected PI association is consistent across genders (GPI effect)

We could consider several models based on what we have learned so far:

- Mutual independence (first-order) model
  - Parameters
    - G: $\beta^G_F(=0), \beta^G_M$—1 parameter estimated, so 1 DF
    - P: $\beta^P_D(=0), \beta^P_R$—1 parameter estimated, so 1 DF
    - I: $\beta^I_{VL}(=0), \beta^I_{SL}, \beta^I_M, \beta^I_{SC}, \beta^I_{VC}$—4 parameter estimated, so 4 DF
  - All GP, GI, and PI ORs are 1. Not likely to be a good fit, because we know that party–ideology association should be important

- Homogeneous association model
  - New parameters:
    - GP: $\beta^{GP}_{FD}(=0), \beta^{GP}_{FR}(=0), \beta^{GP}_{MD}(=0), \beta^{GP}_{MR}$—1 parameter estimated, so 1 DF
    - GI: You figure this out (hint: 10 parameters could be created, but only 4 will actually be estimated)
    - PI: You figure this out
  - Allows GP odds ratio to be $\neq 1$
    - Gender mean ratio in the two parties can be different
    - Equivalently, party mean ratios for two genders can be different
  - Allows PI odds ratios to be $\neq 1$
    - Party (R:D) mean ratio can be different for different ideologies (we know it will)
    - Ideology mean ratios between any pair of ideologies can be different for the two parties
      - e.g., the `VL/VC` ratio can be different for parties `R` and `D`

- Allows GI odds ratios to be $\neq 1$ (you figure out what this means)

- Heterogeneous association model
  - New parameters:
    * GPI: $\beta^{GPI}_{FD1}(=0), \beta^{GPI}_{FR1}(=0), \beta^{GPI}_{MD1}(=0), \beta^{GPI}_{MR1}(=0), \ldots, \beta^{GPI}_{FD5}(=0), \beta^{GPI}_{FR5}(=0), \beta^{GPI}_{MD5}(=0), \beta^{GPI}_{MR5}$
  - Allows GP association ORs to be different for different ideologies
  - Equivalently, allows GI association ORs to be different for different parties
  - Equivalently, allows PI association ORs to be different for different genders

- There are other models we could use. Which one *should* we use??

2.3 Selecting a model

- If we know the correct model, use it.

- More often we have to choose a model using data

- Notice that each order of model is nested within the next one up
  - Can compare orders of model using LR tests.
    * H_a : higher-order model
    * H_0 : lower order model is an adequate fit (extra parameters from higher-order model are all zero)

- Possible strategy:
  1. Compare heterogeneous vs homogeneous association model
     - Difference is three-way interaction
       * Test it using LR test for 3-way interaction
         · `Anova(...)` or `anova(<reduced>, <full>, test = "Chisq")`
     - If LR test is significant, then associations between each pair of variables depends on other variable
       * Use heterogeneous association (saturated) model to estimate ORs for each pair separately at levels of third variable...complicated!
         · `emmeans(...,specs = ~ X + Z + W,...)`
         · `contrast(...,interaction = list("pairwise"), by=X)`
         · `contrast(...,interaction = list("pairwise"), by=Z)`
         · `contrast(...,interaction = list("pairwise"), by=W)`
         · Could use different contrasts in interaction if there are specific questions
     - If LR test is not significant, go to Step 2

2. Compare homogeneous association to mutual independence
   - All two-way interaction terms tested together: Can we dump them all?
   - If significant, then at least one of XZ, XW, or ZW associations is important, but don't know which one(s)
     * Maybe do partial tests on each association term to see if one can be removed
       · The remove least significant term
       · Repeat process on this reduced model
     * Or maybe just keep all two-way interactions
     * Either way, estimate ORs for each pair in the model
       · `emmeans(...,specs = ~ X + Z,...)`
         `contrast(...,interaction = list("pairwise"))`
       · `emmeans(...,specs = ~ X + W,...)`
         `contrast(...,interaction = list("pairwise"))`
       · `emmeans(...,specs = ~ Z + W,...)`
         `contrast(...,interaction = list("pairwise"))`

- **WARNING**: Using data to select model, then using *the same data* to estimate the parameters in the model results in biased parameter estimates
  - You keep only the parameters that "look important" by the selection process
    * These could be type 1 errors
  - You estimate ORs only if data thinks they seem to be needed
    * ORs "seem to be needed" more often when the data overestimates their magnitude
      · If true OR is 1.5, tests are more likely to keep the term if data estimates it as 2.0 than if data estimates it as 1.0!
    * *Therefore, "post-selection" estimates tend to be more extreme than their corresponding population values*
  - On the other hand, if LR test suggests that we can remove an association term, we implicitly estimate ORs to be 1 *with no standard error*
    * Maybe the true value close enough to 1 to make it hard to detect
  - Result is that confidence levels are messed up
    * *Coverage is possibly much less than the started level*
  - INTERPRET RESULTS CAREFULLY AND EXPRESS EXTRA DOUBT IF A CI JUST BARELY EXCLUDES 1
    * Report results as "subject to verification in independent data"

<u>Example</u>: Political Ideology (PolIdeolNominal.R, PolIdeolData.csv)

We will fit loglinear models $Y_{ijk} \sim Po(\mu_{ijk})$, where $i = 1, 2$ represents levels of G, $j = 1, 2$ represents levels of P and $k = 1, 2, 3, 4, 5$ represents levels of I, where different models have different terms in the linear predictor for $\log(\mu_{ijk})$. Reminder of our goals:

1. Determine which associations among GI, GP may be important,

2. Determine whether the expected PI association is consistent across genders (GPI effect)

The strategy we will use is the one outlined above

1. Find a good model

   (a) Fit heterogeneous association and homogeneous association and test the GPI effect (are associations homogeneous across a third factor or not?)

      i. If significant, stop. Final model is heterogeneous association. Go to Step 2

      ii. Otherwise, continue

   (b) Drop GPI and compare homogeneous association and mutual independence.

      i. Compare these models, simultaneously testing GP, GI, and PI

      ii. If not significant, declare no significant associations and estimate all $OR = 1$

      iii. If significant, need at least one interaction term, so test each interaction separately

2. Estimate associations in final model and find VERY APPROXIMATE CIs using `emmeans`

   (a) Final estimates will be biased and CI coverages will be invalid due to reusing same data as was used in model selection, so

First, let's work out some expectations for DF. The numbers of levels for the factors are: G-2, P-2, I-5. Therefore,

- There are $2 \cdot 2 \cdot 5 = 20$ counts we are modeling, so 20 DF total to be accounted for in our models.

- There is 1 intercept

- There are 1, 1, and 4 parameters for main effects of G, P, and I, respectively (DF = levels$-$1)

- There are GP:$1 \cdot 1 = 1$, GI:$1 \cdot 4 = 4$, and PI:$1 \cdot 4 = 4$ parameters in the 2-way interactions

- There are $1 \cdot 1 \cdot 4 = 4$ parameters in the 3-way interaction

Thus, the models have degrees of freedom as follows:

1. Third order: 20 parameters, $20 - 20 = 0$ residual DF (saturated!)

2. Second order: $20 - 4 = 16$ parameters, 4 residual DF

3. First order: $16 - 4 - 4 - 1 = 7$ parameters, 13 residual DF

And for the planned model comparisons:

- Comparing third to second will be a $20 - 16 = 4$ DF test
- Comparing second to first will be a $16 - 7 = 9$ DF test

Now finally, let's do some analysis. Start with comparing (GPI) to (GP, GI, PI) to check for significance of the 3-way interaction.

```
> #Saturated Model:
> mod.sat <- glm(formula = count ~ gender*party*ideol,
+                  family = poisson(link = "log"), data = alldata)
>
> #Homogeneous association model in all 3 associations: GP,GI,PI
> mod.homo <- glm(formula = count ~ (gender + party + ideol)^2,
+                  family = poisson(link = "log"), data = alldata)
> # Test for the G:P:I interaction
> anova(mod.homo, mod.sat, test = "Chisq")
Analysis of Deviance Table

Model 1: count ~ (gender + party + ideol)^2
Model 2: count ~ gender * party * ideol
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1         4     3.2454
2         0     0.0000  4   3.2454   0.5176
```

The test of $H_0 : \log(\mu_{ijk}) = \beta_0 + \beta_i^G + \beta_j^P + \beta_k^I + \beta_{ij}^{GP} + \beta_{ik}^{GI} + \beta_{jk}^{PI}$ vs. $H_a : \log(\mu_{ijk}) = \beta_0 + \beta_i^G + \beta_j^P + \beta_k^I + \beta_{ij}^{GP} + \beta_{ik}^{GI} + \beta_{jk}^{PI} + \beta_{ijk}^{GPI}$ (i.e., $H_0 : \beta_{ijk}^{GPI} = 0$ for all i, j, k) leads to a test statistic $-2\log(\Lambda) = 3.24$ and a p-value of 0.52$>$ 0.05, so we don't reject this H_0. There is not sufficient evidence of a 3-way interaction effect, meaning that we can probably drop this term and use a model that assumes that any 2-way associations act homogeneously over the levels of the third factor. This is a nice simplification, because we can compute and report OR results more simply, not requiring separate versions of each OR for all levels of the third variable.

Next we fit mutual independence and compare it to homogeneous association to see whether we need any interactions at all. We already expect to reject the null hypothesis that all odds ratios are 1, because we believe there should be a strong PI association.

```
> # Mutual Independence
> mod.indep <- glm(formula = count ~ gender + party + ideol,
+                    family = poisson(link = "log"), data = alldata)
> anova(mod.indep, mod.homo, test = "Chisq")
```

```
Analysis of Deviance Table

Model 1: count ~ gender + party + ideol
Model 2: count ~ (gender + party + ideol)^2
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1        13     79.851
2         4      3.245  9   76.606 7.609e-13 ***
```

The test of $H_0 : \log(\mu_{ijk}) = \beta_0+\beta_i^G+\beta_j^P+\beta_k^I+$ vs. $H_a : \log(\mu_{ijk}) = \beta_0+\beta_i^G+\beta_j^P+\beta_k^I+\beta_{ij}^{GP}+\beta_{ik}^{GI}+\beta_{jk}^{PI}$ (what is this in terms of parameters???) leads to a test statistic $-2\log(\Lambda) = 76.6$ and a tiny p-value, so we do reject this H_0. There are clearly some associations in this model. Let's run partial LR tests on each interaction term to see what might be dropped.

```
> library(car)
Loading required package: carData
> Anova(mod.homo)
Analysis of Deviance Table (Type II tests)

Response: count
             LR Chisq Df Pr(>Chisq)
gender         20.637  1  5.551e-06 ***
party           0.528  1    0.46736
ideol         154.131  4  < 2.2e-16 ***
gender:party    3.530  1    0.06026 .
gender:ideol    8.965  4    0.06198 .
party:ideol    60.555  4  2.218e-12 ***
```

Both interactions have p-values of around 0.06, so there is at least a little bit of evidence that each of these interactions might be useful in the model, given that the other one is in. As mentioned above, the type-1 error rates are not reliable for tests performed on models that have already been chosen by the data, as we have done here with (GP, GI, PI). So I am not required to adhere to any strict level of significance for decision making. I choose to keep both of these interactions in the model so that we may better understand their corresponding ORs.

We now study the interactions from these models using `emmeans` ,which is *much* easier than `mcprofile`. I will use the slightly inferior Wald intervals in exchange for being more comfortable that I am doing the analysis right. To see what I mean, the table below shows the parameter estimates from our chosen model. Computing one OR estimate and and corresponding CI requires specifying 16 coefficients—0's 1's, and −1's—in the right order. It is easy to make mistakes. I leave this analysis for the appendix.

```
> round(summary(mod.homo)$coefficients, digits = 3)
               Estimate Std. Error z value Pr(>|z|)
(Intercept)       3.841      0.139  27.673    0.000
```

```
genderM             -0.332      0.197  -1.686    0.092
partyR              -1.105      0.225  -4.909    0.000
ideolSL              0.067      0.190   0.352    0.725
ideolM               0.899      0.162   5.534    0.000
ideolSC             -0.740      0.226  -3.279    0.001
ideolVC             -0.407      0.207  -1.970    0.049
genderM:partyR       0.276      0.147   1.878    0.060
genderM:ideolSL     -0.136      0.265  -0.513    0.608
genderM:ideolM      -0.372      0.227  -1.636    0.102
genderM:ideolSC      0.163      0.269   0.604    0.546
genderM:ideolVC      0.076      0.257   0.297    0.767
partyR:ideolSL       0.424      0.283   1.497    0.134
partyR:ideolM        0.861      0.243   3.548    0.000
partyR:ideolSC       1.687      0.287   5.875    0.000
partyR:ideolVC       1.563      0.273   5.731    0.000
```

Since our associations are homogeneous, we can study ORs by entering the two variables in `specs=` in `emmeans`. This will compute the log-mean at each combination of these two variables averaged across the third variable, which is exactly what we want for studying ORs that are equal across the levels of the third variable. If associations were heterogeneous (different), we would add a `by=` argument with the third variable's name to get the ORs for the two variables in `specs=` computed separately for each level of the variable(s) in `by=`.

The GP association has just 1 parameter, because there are just 2 levels of each factor. So we just need to compute one odds ratio to describe it. We first print out the four estimated values of the linear predictor for the four combinations of `gender` and `party` just so we can see what the function is doing. Then we create the interaction contrast with pairwise comparisons on each factor as `interaction=list("pairwise")`. This is equivalent to asking for `pairwise` ratios (top/bottom) for levels of both variables. If we wanted to reverse the order for, say, `party`, we could use `list("pairwise","revpairwise")`.

```
> emm.GP = emmeans(mod.homo, specs= ~ gender + party)
> # Linear predictors for each combination with unadjusted 95% CIs.
> summary(emm.GP)
 gender party emmean     SE  df asymp.LCL asymp.UCL
 F      D       3.80 0.0692 Inf      3.67      3.94
 M      D       3.42 0.0817 Inf      3.26      3.58
 F      R       3.61 0.0768 Inf      3.46      3.76
 M      R       3.50 0.0804 Inf      3.34      3.65

Results are averaged over the levels of: ideol
Results are given on the log (not the response) scale.
Confidence level used: 0.95
> # Notice that this lists gender first, so it is constructing OR as
> #   F-M ratio for D, divided by F:M ratio for R
```

```
> aa.GP = contrast(emm.GP, interaction=list("pairwise"))
> confint(aa.GP, type="response")
 gender_pairwise party_pairwise ratio    SE  df asymp.LCL
 F / M           D / R           1.32 0.193 Inf     0.988
 asymp.UCL
      1.76

Results are averaged over the levels of: ideol
Confidence level used: 0.95
Intervals are back-transformed from the log scale
```

The function has formulated the OR as

$$OR^{GP}_{FM\backslash DR|k} = \frac{\mu_{FDk}/\mu_{FRk}}{\mu_{MDk}/\mu_{MRk}} = \frac{\mu_{FDk}/\mu_{MDk}}{\mu_{FRk}/\mu_{MRk}}.$$

The estimate is 1.32, so the `D:R` mean ratio is estimated to be 32% higher among females than among males, or equivalently the `F:M` mean ratio is 32% higher in Democrats than in Republicans. The 95% confidence interval is (0.99,1.76), which barely includes 1 as we might expect, given that the LR test suggested near-significance of the effect. Remember that the 95% is just a guess. I would nonetheless conclude that there *may be* a tendency for females to prefer being Democrat vs. Republican more than males do, but this claim requires further evidence to substantiate it.

Looking at the ideologies, we have 5 ordered levels, so there are potentially 10 different pairs of ideologies that we could compare across genders. But the most interesting comparisons (to me) are looking at how the mean ratios of *consecutive* ideologies differ between the genders. We can use the `consec` contrast method for the `ideol` effect in the `interaction=` argument.

```
> emm.GI = emmeans(mod.homo, specs= ~ ideol + gender)
> # Linear predictors for each combination with unadjusted 95% CIs.
> summary(emm.GI)
 ideol gender emmean     SE  df asymp.LCL asymp.UCL
 VL    F        3.29 0.1390 Inf      3.02      3.56
 SL    F        3.57 0.1198 Inf      3.33      3.80
 M     F        4.62 0.0704 Inf      4.48      4.76
 SC    F        3.39 0.1302 Inf      3.14      3.65
 VC    F        3.66 0.1137 Inf      3.44      3.89
 VL    M        3.09 0.1511 Inf      2.80      3.39
 SL    M        3.24 0.1402 Inf      2.96      3.51
 M     M        4.05 0.0933 Inf      3.87      4.23
 SC    M        3.36 0.1327 Inf      3.10      3.62
 VC    M        3.55 0.1208 Inf      3.31      3.78

Results are averaged over the levels of: party
```

```
Results are given on the log (not the response) scale.
Confidence level used: 0.95
> # We want our ORs to be Con:Lib at F vs. Con:Lib at M,
> #   So need to reverse ordering on differences for ideol
> aa.GI = contrast(emm.GI, interaction=list("consec", "pairwise"))
> confint(aa.GI, type="response", adjust="none")
 ideol_consec gender_pairwise ratio    SE  df asymp.LCL
 SL / VL      F / M           1.145 0.303 Inf     0.682
 M / SL       F / M           1.266 0.273 Inf     0.829
 SC / M       F / M           0.586 0.126 Inf     0.384
 VC / SC      F / M           1.090 0.263 Inf     0.679
 asymp.UCL
     1.924
     1.933
     0.895
     1.750

Results are averaged over the levels of: party
Confidence level used: 0.95
Intervals are back-transformed from the log scale
```

The ideologies are listed in the order we specified earlier, from `VL` to `VC`. The `consec` method compares lower to higher levels the order, so ratios of a more conservative to a more liberal ideology for females against males. Any OR above 1 indicates that females have more conservative ideology than males. Interestingly, three of these ORs are relatively close to 1, with CIs that put 1 pretty far from the boundaries. But one of them, `SC:M`, has an estimate a bit farther from 1 and a CI that is not very close to 1. So it seems that females are somewhat less likely to choose slightly conservative over moderate compared to males.

---

3 Four or more variables

Nothing we have done requires limiting the problem to 3 variables

- p categorical variables form a *p-way table*
  - Can no longer depict the table in our 3D world
- Goal is to learn about *associations* among variables
  - Which *pairs* are associated, which are not?
  - Are associations between two variables consistent across levels of other variables, or do they change?
    - i.e., does the value of a particular odds ratio depend on levels of another variable?

- Models focus on understanding two-way interactions and how they might change for levels of other factors
  * Multiple possible three-way interactions
  * Four-way interaction means that the three-way interaction changes across levels of the 4th variable
    - *My brain hurts!*

4 Conclusions: What to learn

1. How to think about modeling 3 or more variables
2. How to construct the models
   (a) Terms, parameters in the model,
   (b) Comparisons of models using LR tests
3. How to build models in R
4. How to create ORs for 2-variable associations with or without a third (or a fourth!) variable involved.

5 Appendix: LR Analysis using `mcprofile`

The GP association has just 1 parameter, because there are just 2 levels of each factor. So we just need to compute one odds ratio to describe it, because the homogeneous association model says that the GP association is the same for all levels of `ideol`. The levels of G are `F` and `M` in that order, and the levels of P are `D` and `R` in that order. Thus, if we ask whether the mean ratio of Democrats to Republicans is the same for females and males, this leads to:

$$OR^{GP}_{FM,DR(k)} = \frac{\mu_{FDk}/\mu_{FRk}}{\mu_{MDk}/\mu_{MRk}} = \frac{\mu_{FDk}\mu_{MRk}}{\mu_{MDk}\mu_{FRk}} = \exp(\beta^{GP}_{FD} + \beta^{GP}_{MR} - \beta^{GP}_{FR} - \beta^{GP}_{MD}).$$

Notice that only one of these parameters actually exists in the estimated model (β^{GP}_{MR}, as represented by the `genderM:partyR` parameter in the output); the rest are 0. Thus, we estimate $OR^{GP}_{FM,DR(k)}$ using a coefficient vector,

$$0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0$$

```
> contr.mat.GP <- rbind(c(rep(0, 7), 1,rep(0, 8)))
> row.names(contr.mat.GP) <- c("GP Dem/Rep:F/M")
> library(mcprofile)
> LRCI <- mcprofile(mod.homo, CM = contr.mat.GP)
> exp(confint(LRCI, adjust = "none"))

  mcprofile - Confidence Intervals

level:          0.95
adjustment:     none

                 Estimate lower upper
GP Dem/Rep:F/M       1.32 0.988  1.76
```

The estimated odds ratio is 1.32, meaning that the ratio of Democrats to Republicans is 32% higher for females than for males. The 95% LR confidence interval is 0.99–1.76, so the CI just barely includes 1. This indicates that the GP association is not *quite* significant at the 0.05 level. I would nonetheless conclude that there *may be* a tendency for females to prefer being Democrat vs. Republican more than males do, but this claim requires further evidence to substantiate it.

Looking at the ideologies, we have 5 levels, so there are potentially 10 different pairs of ideologies that we could compare across genders. We can do all 10 without difficulty (I show them below), but in practice I would perhaps try to get this list narrowed down to the most meaningful comparisons according to the subject-matter specialist. Note that the order of the I levels is `VL`, `SL`, `M`, `SC`, `VC`, and that these levels are ordered. We could create mean ratios such that the numerator is the more liberal ideology and the denominator is the *next* more conservative one, and compare these ideology ratios for females to males. For example, fixing `party` at any level j, we would compare $(\mu_{F,j,SC}/\mu_{F,j,VC})$ to $(\mu_{M,j,SC}/\mu_{M,j,VC})$ using an odds ratio:

$$OR^{GI}_{FM;(j);VC,SC} = \frac{\mu_{F,j,VC}\mu_{M,j,SC}}{\mu_{F,j,SC}\mu_{M,j,VC}} = \exp(\beta^{GI}_{F,VC} + \beta^{GI}_{M,SC} - \beta^{GI}_{F,SC} - \beta^{GI}_{M,VC}).$$

Note here that the two female parameters don't exist, and the first 8 and last 4 parameters overall are unrelated to the GI interaction. Thus, the coefficient vector for this OR is as follows:

$$0, 0, 0, 0, 0, 0, 0, 0, \quad 0, 0, 1, -1, \quad 0, 0, 0, 0.$$

The entire coefficient matrix for all 10 ORs is as shown in the output below.

```
> contr.mat.GI <- rbind(
+   c(rep(0, 8), 0, 0, 1,-1, rep(0, 4)),
+   c(rep(0, 8), 0, 1, 0,-1, rep(0, 4)),
+   c(rep(0, 8), 1, 0, 0,-1, rep(0, 4)),
+   c(rep(0, 8), 0, 0, 0,-1, rep(0, 4)),
+   c(rep(0, 8), 0, 1,-1, 0, rep(0, 4)),
+   c(rep(0, 8), 1, 0,-1, 0, rep(0, 4)),
+   c(rep(0, 8), 0, 0,-1, 0, rep(0, 4)),
+   c(rep(0, 8), 1,-1, 0, 0, rep(0, 4)),
+   c(rep(0, 8), 0,-1, 0, 0, rep(0, 4)),
+   c(rep(0, 8),-1, 0, 0, 0, rep(0, 4)))
> row.names(contr.mat.GI) <-
+   c("GI VC:SC | F:M", "GI VC:M | F:M", "GI VC:SL | F:M",
+     "GI VC:VL | F:M", "GI SC:M | F:M", "GI SC:SL | F:M",
+     "GI SC:VL | F:M", "GI M:SL | F:M", "GI M:VL | F:M",
+     "GI SL:VL | F:M")
> LRCI <- mcprofile(mod.homo, CM = contr.mat.GI)
> exp(confint(LRCI, adjust = "none"))

  mcprofile - Confidence Intervals

level:           0.95
adjustment:      none

                 Estimate lower upper
GI VC:SC | F:M      1.090 0.679 1.751
GI VC:M | F:M       0.639 0.431 0.947
GI VC:SL | F:M      0.809 0.499 1.309
GI VC:VL | F:M      0.927 0.559 1.533
GI SC:M | F:M       0.586 0.383 0.895
GI SC:SL | F:M      0.742 0.446 1.230
GI SC:VL | F:M      0.850 0.500 1.441
GI M:SL | F:M       1.266 0.827 1.931
```

```
GI M:VL | F:M      1.450 0.927 2.263
GI SL:VL | F:M     1.145 0.682 1.926
```

It's a little tricky to try to interpret all of this. First, note that only two CIs exclude 1 as a possible value, $OR^{GI}_{FM;(j);VC,M}$ and $OR^{GI}_{FM;(j);SC,M}$. Notice that both of these involve comparisons of a conservative to a moderate (central) ideology. The estimates are 0.64 and 0.59, respectively, indicating that females are less likely to choose the very conservative or slightly conservative ideology over moderate ideology than males are. I notice that the other comparisons with moderate ideology are 1.27 and 1.45 for the moderate-to-liberal ratios. Again, this indicates that females are more likely to choose the moderate ideology over over a liberal ideology than males are, although the evidence in these cases is not statistically significant at the 0.05 level. All of these cases point to a tendency for females to be more moderate than males.

6 Exercises (due when announced)

In an early paper on the relationship between smoking and early death, Hammond and Horn (1954) study past smoking status and death rates of 187,766 men over a 20-month period. The data are given in the file `JAMA1954Smoking.dat`. The variables are `Smoker` (=`0` if very little or no smoking in their lifetimes, =`1` otherwise), `Agecat` (=`1` for men of ages 50–54 at the start of the study, =`2` for 55–59, =`3` for 60–64, and =`4` for ages 65–69), `Death` (=`1` if the subject died during the 20-month follow-up period, =`0` otherwise), and `Count` of the number of men in each combination of the other three variables.

The main question is whether smoking and death are associated. A secondary interest is in whether the association between smoking and death changes across age groups. If it does, then how, then how is it different?

1. Using symbols S for `Smoker`, A for `Agecat`, and D for `Death`, write out the full distributional model for the counts in abbreviated notation for each term ($\beta_i^X + \beta_j^Y + ...$for $i = 1, 2;j = 1, 2, 3...$)
   (a) How many parameters will be estimated for each term in the model?
2. Fit a saturated loglinear model to these data.
   (a) Check the `summary()` to ensure that the correct number of parameters is estimated for each term.
   (b) Perform LR tests on all terms in the model. Report test statistics and p-values.
   (c) Consider the test for the three-way interaction. Write out the hypotheses in terms of model parameters and in terms of a model comparison. Interpret the test results. Specifically, what question about smoking and death does this test attempt to answer for the researchers?
3. Since there is a significant 3-way interaction, we explore the relationship between smoking and death separately for each age group.
   (a) Compute the ratio of the odds of death for smokers to the odds of death for non-smokers for each age category along with 95% Wald confidence intervals.
   (b) Interpret the results. Specifically, answer these questions
      i. Does smoking appear to be associated with increased risk of death in each age group?
      ii. Is there an apparent pattern in the strength of this association across age groups?

STATISTICS 475: Applied Discrete Data Analysis

Lecture 23: Loglinear models with Ordinal Variables

(B&L Section 4.2.6)

1 Problem to be solved

- Generally speaking, **statistical models try to mimic structures in a population**
  - Models try to approximate structure in a population
    - Linearity, presence of interactions, restrictions on parameter values, etc.
  - When they are close to right, they give us much more stable and reliable estimates and inferences than model-less analysis
    - Linear regression is better than zig-zag interpolation when straight line is reasonable approximation
  - The more we know about a population, the more details we can add to our model to gain these benefits
- Ordinal categorical variables are very common!
  - Political ideology: is ordinal:
    - 1: "Very Liberal" (`"VL"`), 2: "Slightly Liberal" (`"SL"`), 3: "Moderate" (`"M"`), 4: "Slightly Conservative" (`"SC"`), and 5: "Very Conservative" (`"VC"`)
  - There is some real or conceptual scale that can identify levels as "more" and "less"
- So far, our loglinear models have been constructed without taking advantage of information about potential ordering of categories within a variable
- Need to develop methods for handling ordinal variables in loglinear models
  - How is ordinality brought into a model?
  - How what special questions can the structure answer?
  - What are its limitations?

2 General Approach to Ordinal Variables

- Loglinear models are all about modeling *associations* between pairs of variables
- *Associations are measured by comparing ratios of means between two levels of one variable across levels of another*
  - Need to imagine what ratios of means look like *within* an ordinal variable
  - Need to imagine how ratios of means might change across levels of an *ordinal* variable
    - Maybe they change in a specific pattern relating to the ordering, rather than arbitrarily as with nominal variables
- Presently, our nominal loglinear model does not impose any special relationships on the mean ratios among different ideologies.
  - For example, the model allows the ratio of `party` means within a given ideology—for example the ratio of Republicans to Democrats, say μ_{iRk}/μ_{iDk}—to change in any arbitrary manner as we change ideologies.
- However, we know that Democrats tend to be more liberal than Republicans.
  - Thus, we strongly suspect that this mean ratio would change systematically, in a pattern, as we move across the spectrum from $k = 1$ (`VL`) to $k = 5$ (`VC`).
  - Specifically, we might expect that

$$\frac{\mu_{iR1}}{\mu_{iD1}} < \frac{\mu_{iR2}}{\mu_{iD2}} < \frac{\mu_{iR3}}{\mu_{iD3}} < \frac{\mu_{iR4}}{\mu_{iD4}} < \frac{\mu_{iR5}}{\mu_{iD5}}$$

- How can we incorporate this feature into a *model*, test whether it is reasonable, and use the model to compute better estimates of ORs?
- In a loglinear model, associations are modeled as interactions.
- The main approach to modeling association involving ordinal variables is to assign numeric SCORES to the levels and treat them as numerical in *interactions* involving that variable
  - Our models match the marginal totals perfectly and try to understand structure of interaction terms.
- The main new challenges to modeling Nominal and ordinal categorical variables together are:
  1. Choosing appropriate scores (I handle this last)
  2. Knowing where to use scores in model
  3. Interpreting parameters
  4. Developing estimates, mean ratios, and odds ratios
  5. Checking whether scores "fit" the data

3 Choosing scores

What numbers should we assign to levels of a variable?

- If we have J levels of an ordinal variable, we need a numerical variable, say s, with J values, the scores, say $s_1, s_2, \ldots, s_J$.
- Scores are subjective, there are no formal rules governing the choices
  - Subject-matter specialist should help
- Mainly need them to reflect ordering on whatever scale you want them to measure
- Useful detail: Models we will create will depend on scores only through their *relative spacings*, and not through their actual values
  - Two sets of scores with equal spacings yield the same estimated ORs, CIs, and test results
    - 1-2-3 = 10-11-12
  - Two sets of scores with equal *proportional* spacings yield the same tests but not estimates
    - 1-2-4-8 = 10-20-40-80 for tests because gaps are always in 1:2:4 ratio
  - Different directions (1-2-3 vs. 3-2-1) give different definitions to what is "high" or "low", but otherwise give the same tests but with reversed ORs
    - `ideol`: doesn't matter whether you use 1-2-3-4-5 or 5-4-3-2-1.
    - In first set, higher score is more conservative; in second set higher score is more liberal
    - In both sets, a "1-unit change" moves from one category to its neighbour, just in opposite directions
      - Therefore a 1-unit change in 1-2-3-4-5 gives exactly the same mean ratio as a -1 unit change in 5-4-3-2-1
      - The mean ratio from a 1 unit change in 1-2-3-4-5 is the inverse of the ratio from a 1-unit change in 5-4-3-2-1
- *Convenient if scale should be chosen to make a 1-unit change in score meaningful*
  - You get to decide this!
  - "Default" is to use consecutive integers, so that 1-unit increase is one-category change (1-2-3-4-5)
    - Convenient, but implies that changes between all consecutive categories are all equivalent
    - Might makes sense to make the smallest gap 1.
    - 10-20-30-40-50 is probably not meaningful: a 1-unit change is 1/10 of a category...what does this mean?

    - Special cases exist
        - For age groups like “<25”, “25-45”, and “>45” we might use scores like 20-35-60
        - Then a 1-unit change in score is meant to roughly represent 1-year change in age.
  - Can split some pairs farther apart if you think certain categories are more different from neighbours
    - For example, might think that ideology difference between “very...” and “slightly...” is bigger than difference between “slightly” and “moderate”.
    - Use 1-3-4-5-7 (or 0-2-3-4-6) scores

- *Can design different scores to reflect different conceptual scales!*
  - 1-2-3-4-5 represents increasing conservativeness with increasing scores
  - 2-1-0-1-2 represents increasing *extremeness* with increasing scores
    - (Yes, it's OK to assign equal scores it two categories are equivalent on some scale)

- If you use do multiple analyses of the same data set, you don't get to pick the one you like best and pretend the others didn't exist
  - Report all analyses performed, e.g., trying our different sets of scores for increasing conservativeness
  - If they give similar results, then report either one and mention that the others support the results
  - If they give different results, then *report both* and indicate that there is some uncertainty regarding the conclusions.
    - YOU didn't know which scores were better, so you shouldn't now pretend that you did just because you like one set of results better.

4 Models

We can fit different models depending on whether we have one or more than one ordinal variable. The different cases are described below. In all cases, model need to

1. Match marginal counts perfectly

2. Allow mean ratios to change in some pattern according to the scores.

4.1 Two categorical variables, one is ordinal

- Consider two variables, X with I levels and Z with J levels, and suppose Z is ordinal.
  - To help with ideas, let $I = 2$:

μ_{ij}		Z				
		1	2	3	4	Total
X	1	μ_{11}	μ_{12}	μ_{13}	μ_{14}	
	2	μ_{21}	μ_{22}	μ_{23}	μ_{24}	
	Total					

- Remember that main effects in models for tables of counts serve only to adjust *marginal counts for that variable* to match observed counts.
  - Without the main effects, interaction terms contain a bizarre combination of association between variables and popularity of individual response levels
    - Makes interpretation a mess.
  - So ALL models start from *nominal* main effects (indicators for each category) and move on from there
  - Interaction parameters impact mean ratios, and hence odds ratios that measure associations
- Recall from a nominal model, $\log(\mu_{ij}) = \beta_0 + \beta_i^X + \beta_j^Z + \beta_{ij}^{XZ}$
  - In our example, ratios of means in the same column are

$$\frac{\mu_{1j}}{\mu_{2j}} = \exp([\beta_1^X - \beta_2^X] + [\beta_{1j}^{XZ} - \beta_{2j}^{XZ}])$$

    and ratios of means in the same row are

$$\frac{\mu_{ia}}{\mu_{ib}} = \exp([\beta_a^Z - \beta_b^Z] + [\beta_{ia}^{XZ} - \beta_{ib}^{XZ}])$$

  - Odds ratios are just ratios of pairs of these:

$$\frac{\mu_{1a}/\mu_{2a}}{\mu_{1b}/\mu_{2b}} = \frac{\mu_{1a}/\mu_{1b}}{\mu_{2a}/\mu_{2b}} = \frac{\mu_{1a} * \mu_{2b}}{\mu_{2a} * \mu_{1b}} = \exp(\beta_{1a}^{XY} + \beta_{2b}^{XY} - \beta_{2a}^{XY} - \beta_{1b}^{XY}).$$

  - These can be anything in the world, depending on the values of the parameters
    - Not constrained to follow a pattern as levels of a and b change
- *To model ordinal associations, put scores in interaction*
  - Instead of putting indicators $z_1, z_2, \ldots, z_J$ for Z in crossproducts, use score variable s
    - 2×3 Nominal model in regression form, including "disappearing terms":

$$\log(\mu_{ij}) = \ldots + \beta_{11}^{XZ} x_1 z_1 + \beta_{12}^{XZ} x_1 z_2 + \beta_{13}^{XZ} x_1 z_3 + \beta_{21}^{XZ} x_2 z_1 + \beta_{22}^{XZ} x_2 z_2 + \beta_{23}^{XZ} x_2 z_3$$

    - 2×3 **Ordinal** model in regression form:

$$\log(\mu_{ij}) = \ldots + \beta_1^{XZ} x_1 s + \beta_2^{XZ} x_2 s$$

    * In abbreviated form, can say $\log(\mu_{ij}) = \beta_0 + \beta_i^X + \beta_j^Z + \beta_i^{XZ} s_j$

- Simpler model imposes structure on mean ratios and odds ratios:
  - Ratios of means in the same column are found from

$$\log\left(\frac{\mu_{1j}}{\mu_{2j}}\right) = [\beta_1^X - \beta_2^X] + [\beta_1^{XZ} - \beta_2^{XZ}]s_j$$

    * *These ratios change linearly as the scores change*
  - Ratios of means in the same row are found from

$$\log\left(\frac{\mu_{ia}}{\mu_{ib}}\right) = [\beta_a^Z - \beta_b^Z] + \beta_i^{XZ}[s_a - s_b]$$

    * *These ratios are the same for any pair of columns with the same difference in scores*
  - Odds ratios are just ratios of pairs of these:

$$\log\left(\frac{\mu_{1a}/\mu_{2a}}{\mu_{1b}/\mu_{2b}}\right) = \log\left(\frac{\mu_{1a}/\mu_{1b}}{\mu_{2a}/\mu_{2b}}\right) = \log\left(\frac{\mu_{1a} * \mu_{2b}}{\mu_{2a} * \mu_{1b}}\right) = [\beta_1^{XY} - \beta_2^{XY}][s_a - s_b].$$

    * *These are also the same for any pair of Z levels with the same difference in scores*

- As a result of this dependence on the linear difference in scores, this model is called the Linear Association model
  - Odds ratios involving columns that are 1 unit apart in s are identical
  - Odds ratios for column 2 units apart in s are the squared of those 1 unit apart
    * More separation just increases the power on the OR
    * If $OR(1)$ is the OR for a 1-unit spacing, then for a c-unit spacing, $OR(c) = OR(1)^c$

- Degrees of freedom for $\log(\mu_{ij}) = \beta_0 + \beta_i^X + \beta_j^Z + \beta_i^{XZ} s_j$, $i = 1, \ldots, I$; $j = 1, \ldots, J$
  - Underlying regression has
    * $I - 1$ parameters for X main effects β_i^X, $i = 1, \ldots, I$ (first parameter is 0)
    * $J - 1$ parameters for Z main effects β_j^Z, $j = 1, \ldots, J$ (first parameter is 0)
    * ***$I - 1$ parameters for XZ interaction (association) parameters β_i^{XZ}, $i = 1, \ldots, I$ (first parameter is 0)
  - Compare to nominal association (saturated) model with $(I-1)(J-1)$ interaction parameters
    * Difference is $(I - 1)(J - 2)$ DF

- In summary, the analysis of a nominal $\times$ ordinal table consists of:

– Creating scores for the ordinal variable Z (call the variable `scores` for now)

– Fitting Poisson model: `glm(y~X + Z + X*scores)`

  * Note: `glm()` will automatically add a useless main effect for the `scores` variable. See example below.

– Testing the significance of association: `Anova()`

– Examining means ratios and/or ORs for two rows of X as Z-scores change by c units:

  * Need to do a special "trick" to make `emmeans()` work, but it's not hard.
  * `mcprofile()` *will not work* on these models

- Linear association is a special case of nominal association where we force mean ratios into a special pattern

  – Can do LR test to compare linear association model to nominal association model to see if the model for the log-ORs is an adequate fit

  – H_0 : Linear association adequately describes association in data

    * $H_0 : \log(\mu_{ij}) = \beta_0 + \beta_i^X + \beta_j^Z + \beta_i^{XZ} s_j$, $i = 1, \ldots, I$; $j = 1, \ldots, J$

  – H_a : Linear association does not adequately describe association in data

    * $H_0 : \log(\mu_{ij}) = \beta_0 + \beta_i^X + \beta_j^Z + \beta_{ij}^{XZ}$, $i = 1, \ldots, I$; $j = 1, \ldots, J$

<u>Example</u>: Political Ideology (Lecture 23 scripts.R, PolIdeolData.csv)

(Mostly from book) Our work with nominal models has suggested that the three-variable interaction is not needed, so we start with a homogeneous association model and consider adding scores. To begin, we consider 1-2-3-4-5 scores contained in a new variable we call `lin.score.1`, where higher scores mean greater conservativeness. We apply the scores to the PI term, because we believe that the odds ratios from this association should become more extreme as the difference between ideologies increases. We will leave everything else as it was before, in particular allowing the association between gender and ideology to remain nominal, so we can focus on what happens with this single change to the model.

Thus, the model we fit is

$$\log(\mu_{ijk}) \quad = \quad \beta_0 + \beta_i^G + \beta_j^P + \beta_k^I + \beta_{ij}^{GP} + \beta_{ik}^{GI} + \beta_j^{PI} s_k^I, \tag{1}$$

which assumes linear association between party and ideology that is homogeneous across genders. The model assumes that a comparison of the ratio of Republicans to Democrats at ideology levels k_1 and k_2 has the form

$$\log(OR_{RD,k_1k_2}^{PI}) \quad = \quad (\beta_R^{PI} - \beta_D^{PI})(s_{k_1}^I - s_{k_2}^I)$$

for both genders, where the value of $s_{k_1}^I - s_{k_2}^I$ is the difference in scores between the two ideologies being compared. Notice that using consecutive integer scores (1-2-3-4-5) implies that the odds ratios involving any two consecutive ideologies are the same, since the difference is always 1 (or -1, depending on the direction in which ratios are formed). We fit and analyze this linear PI-association model in detail below.

Model Fit The terms in Equation 1 are specified concisely in `glm()` by using only the interaction terms as shown below:

```
> mod.homo.lin1.PI <- glm(formula = count ~ gender*party +
                          gender*ideol + party*lin.score1,
                          family = poisson(link = "log"),
                          data = alldata)
> summary(mod.homo.lin1.PI)

Call: glm(formula = count ~ gender * party + gender * ideol +
          party * lin.score1, family = poisson(link = "log"),
          data = alldata)

Deviance Residuals:
     Min        1Q    Median        3Q       Max
-1.24363  -0.54155   0.02178   0.52057   0.90547

Coefficients: (1 not defined because of singularities)
                  Estimate Std. Error z value Pr(>|z|)
(Intercept)        3.83483    0.13312  28.807  < 2e-16 ***
genderM           -0.33690    0.19696  -1.711  0.08717 .
partyR            -1.51082    0.20862  -7.242 4.42e-13 ***
ideolSL            0.06249    0.17211   0.363  0.71654
ideolM             0.89370    0.14906   5.996 2.03e-09 ***
ideolSC           -0.51217    0.19204  -2.667  0.00765 **
ideolVC           -0.51766    0.20362  -2.542  0.01101 *
lin.score1              NA         NA      NA       NA
genderM:partyR     0.28723    0.14555   1.973  0.04845 *
genderM:ideolSL   -0.13744    0.26405  -0.520  0.60273
genderM:ideolM    -0.37450    0.22682  -1.651  0.09872 .
genderM:ideolSC    0.18409    0.26655   0.691  0.48979
genderM:ideolVC    0.06172    0.25808   0.239  0.81099
partyR:lin.score1  0.43058    0.06005   7.170 7.48e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 255.1481  on 19  degrees of freedom
Residual deviance:   8.3986  on  7  degrees of freedom
AIC: 142.83

Number of Fisher Scoring iterations: 4
```

Because the linear score `lin.score1` appears only in an interaction and not as a main effect in the `formula` argument, a main effect is automatically created for it by `glm()`. Similarly, main effects are created for `gender`, `party`, and `ideol`. This means that the main effect of ideology is included into the model both as a series of indicator variables and as a numerical variable. This duplication leads to the note about "singularities" in the listing of estimated model coefficients and the "`NA`" results for the `lin.score1` main effect.

This redundancy and resulting lack of estimate creates no difficulty in interpretation, because we are not modeling the main effects of ideology score. The parameter for the interaction between `party` and `lin.score1` is properly estimated, so we can mostly ignore the issue, save for some computational details noted below and in the program. We can coerce `emmeans()` to give Wald inferences through an extra statement that tells it to equate the 5 levels of `lin.score1` to the corresponding 5 levels of `ideol`.

We start with the tests for effects using `Anova()`.

```
> library(car)
> Anova(mod.homo.lin1.PI)
Analysis of Deviance Table (Type II tests)
Response: count
                 LR Chisq Df Pr(>Chisq)
gender             20.637  1  5.551e-06 ***
party               0.528  1    0.46736
ideol             163.007  3  < 2.2e-16 ***
lin.score1                 0
gender:party        3.901  1    0.04826 *
gender:ideol        9.336  4    0.05323 .
party:lin.score1   55.402  1  9.825e-14 ***
```

We see that there is a very strong linear association between party and ideology, which we already knew, but now the LR test has just 1 df. This tells us that there is a strong *linear* association between party and ideology. Tests for the other associations yield results very similar to those in the nominal case, although if one uses a rigid 0.05 significance level, GP is now significant. Practically speaking, the two tests are telling us the same thing: there is borderline evidence to suggest that GP and GI interactions are present, given the other terms in the model.

Analysis of means and ORs Moving to means and odds ratios, the superfluous `lin.score1` term in the model messes up all procedures we have for follow-up calculations. However, `emmeans` is relatively easily adapted to run as expected. We simply need to tell it to treat the 5 scores as equivalent to the 5 levels of `ideol` using the `ref_grid()` function. The `cov.reduce = A~B` argument equates levels of the score variable `A` to levels of its nominal form `B`. The results below show that the mapping of numeric scores to factor levels has worked.

```
> rg1 = ref_grid(mod.homo.lin1.PI, cov.reduce = lin.score1 ~ ideol)
> summary(rg1)
```

```
 gender party ideol lin.score1 prediction     SE  df
 F      D     VL             1       3.83 0.1331 Inf
 M      D     VL             1       3.50 0.1537 Inf
 F      R     VL             1       2.75 0.1730 Inf
 M      R     VL             1       2.70 0.1858 Inf
 F      D     SL             2       3.90 0.1220 Inf
 M      D     SL             2       3.42 0.1486 Inf
 F      R     SL             2       3.25 0.1380 Inf
 M      R     SL             2       3.06 0.1586 Inf
 F      D     M              3       4.73 0.0817 Inf
 M      D     M              3       4.02 0.1095 Inf
 F      R     M              3       4.51 0.0875 Inf
 M      R     M              3       4.09 0.1076 Inf
 F      D     SC             4       3.32 0.1405 Inf
 M      D     SC             4       3.17 0.1471 Inf
 F      R     SC             4       3.53 0.1360 Inf
 M      R     SC             4       3.67 0.1342 Inf
 F      D     VC             5       3.32 0.1480 Inf
 M      D     VC             5       3.04 0.1619 Inf
 F      R     VC             5       3.96 0.1229 Inf
 M      R     VC             5       3.97 0.1245 Inf

Results are given on the log (not the response) scale.
```

Now we use the `emmeans()` function as we have before, using the new "regridded" object in place of the model. Here are the model-estimated means for each combination of the three factors.

```
> emm1a = emmeans(rg1, specs=~ ideol + party + gender)
> aa1a = confint(emm1a, type="response")
> aa1a
 ideol party gender  rate   SE  df asymp.LCL asymp.UCL
 VL    D     F       46.3 6.16 Inf      35.7      60.1
 SL    D     F       49.3 6.01 Inf      38.8      62.6
 M     D     F      113.1 9.24 Inf      96.4     132.8
 SC    D     F       27.7 3.90 Inf      21.1      36.5
 VC    D     F       27.6 4.08 Inf      20.6      36.9
 VL    R     F       15.7 2.72 Inf      11.2      22.1
 SL    R     F       25.7 3.55 Inf      19.6      33.7
 M     R     F       90.9 7.95 Inf      76.6     107.9
 SC    R     F       34.3 4.66 Inf      26.2      44.7
 VC    R     F       52.4 6.44 Inf      41.2      66.7
 VL    D     M       33.0 5.08 Inf      24.4      44.7
 SL    D     M       30.7 4.56 Inf      22.9      41.0
```

```
 M      D      M         55.5  6.08  Inf        44.8        68.8
 SC     D      M         23.8  3.50  Inf        17.8        31.8
 VC     D      M         20.9  3.39  Inf        15.3        28.8
 VL     R      M         15.0  2.78  Inf        10.4        21.5
 SL     R      M         21.3  3.38  Inf        15.6        29.1
 M      R      M         59.5  6.40  Inf        48.2        73.4
 SC     R      M         39.2  5.26  Inf        30.1        51.0
 VC     R      M         53.1  6.60  Inf        41.6        67.7

Confidence level used: 0.95
Intervals are back-transformed from the log scale
```

Odds ratio computations for the nominal GP and GI associations are exactly as before. These calculations are in the program and we do not repeat them here.

For PI linear association, we can use `emmeans` average the linear predictors across levels of the third, unneeded, factor.

```
> (emm.PI1 = emmeans(rg1, ~ ideol + party))
 ideol party emmean     SE  df asymp.LCL asymp.UCL
 VL    D       3.67 0.1048 Inf      3.46      3.87
 SL    D       3.66 0.0978 Inf      3.47      3.85
 M     D       4.37 0.0684 Inf      4.24      4.51
 SC    D       3.25 0.1041 Inf      3.04      3.45
 VC    D       3.18 0.1226 Inf      2.94      3.42
 VL    R       2.73 0.1424 Inf      2.45      3.01
 SL    R       3.15 0.1092 Inf      2.94      3.37
 M     R       4.30 0.0694 Inf      4.16      4.43
 SC    R       3.60 0.0967 Inf      3.41      3.79
 VC    R       3.97 0.0909 Inf      3.79      4.14

Results are averaged over the levels of: gender
Results are given on the log (not the response) scale.
Confidence level used: 0.95
```

Then we compute the PI log-OR,

$$\log(OR^{PI}_{RD,k_1k_2}) \;=\; (\beta^{PI}_R - \beta^{PI}_D)(s^I_{k_1} - s^I_{k_2}),$$

remembering that we will have $\beta^{PI}_D = 0$. Finally, we compute ORs using consecutive levels of ideology. Since we used scores of 1-2-3-4-5, we expect that ORs based on comparing the R:D ratio for consecutive levels of increasing conservativeness should all involve a 1-unit change in s, and hence have identical ORs. We see this below:

```
> aa.PI = contrast(emm.PI1, interaction="consec")
> confint(aa.PI, type="response")
 ideol_consec party_consec ratio     SE  df asymp.LCL asymp.UCL
 SL / VL      R / D         1.54 0.0924 Inf      1.37
1.73
 M / SL       R / D         1.54 0.0924 Inf      1.37
1.73
 SC / M       R / D         1.54 0.0924 Inf      1.37
1.73
 VC / SC      R / D         1.54 0.0924 Inf      1.37
1.73

Results are averaged over the levels of: gender
Confidence level used: 0.95
Intervals are back-transformed from the log scale
```

There is about a 54% increase in the R:D mean ratio as one moves from a more liberal category into the next more conservative category, and the CI ranges from 37% to 73%.

We can further look at ORs corresponding to all possible differences of scores—1, 2, 3, and 4—simply by comparing all other levels against `VL`, the first level. We can do this using the `trt.vs.ctrl1` contrast method. Here we want `VL`, the first level, to be the baseline that we compare to, so we use `trt.vs.ctrl1`.

```
> aa.PI2 = contrast(emm.PI1,
interaction=list("trt.vs.ctrl1", "consec"))
> confint(aa.PI2, type="response")
 ideol_trt.vs.ctrl1 party_consec ratio     SE  df asymp.LCL asymp.UCL
 SL / VL            R / D         1.54 0.0924 Inf      1.37
1.73
 M / VL             R / D         2.37 0.2841 Inf      1.87
2.99
 SC / VL            R / D         3.64 0.6556 Inf      2.56
5.18
 VC / VL            R / D         5.60 1.3445 Inf      3.50
8.96

Results are averaged over the levels of: gender
Confidence level used: 0.95
Intervals are back-transformed from the log scale
```

You can verify that these odds ratios are all powers of 1.54. For adjacent categories of ideology, the R:D mean ratio (or the odds of being Republican) is 1.54 times as high for the more conservative ideology than for the more liberal one ($1.37 < OR < 1.73$). For a 2-category difference, the R:D ratio increase to 2.37 times as high in the more conservative ideology ($1.87 < OR < 2.99$); for 3 categories to 3.64 ($2.56 < OR < 5.18$); and finally the odds of being Republican are 5.6 times as high ($3.50 < OR < 8.96$) for very conservative ideology compared to very liberal ideology. While the numbers themselves may not hold special meaning to us, the pattern is exactly what we would expect.

Comparison to nominal model When viewed nominally, the R:D odds ratio estimates based on a 1-unit increase in conservatism consecutive ideologies were 0.88, 2.28, 1.55, and 1.53. We now have replaced these with one estimate, 1.54, which lise somewhat in the middle of these estimates. However, in the nominal analysis, the first two of these estimates (`VC:SC` and `SC:M`) had confidence intervals that did not overlap (0.54 to 1.46 for `VC:SC`, and 1.48 to 3.52 for `SC:M`). This suggests that the true odds ratios that these two intervals are estimating may not both have the same value, contrary to the assumption that we make when we assign 1-2-3-4-5 scores. Thus, it remains to be seen whether the linear pattern for $\log(OR)$ really fits these data.

We check the fit of the linear-association assumption with a LR test. The nominal association model is the full model used for the alternative hypothesis, and the linear association model is the reduced model for the null. Thus, equivalently, we have H_0 : linear association is true in these data vs. H_a : linear association is not true in these data. We do the model comparison through `anova(H0, Ha)`:

```
> mod.homo <- glm(formula = count ~ (gender + party + ideol)^2,
                  family = poisson(link = "log"), data = alldata)
> anova(mod.homo.lin1.PI, mod.homo, test = "Chisq")
Analysis of Deviance Table

Model 1:count ~ gender * party + gender * ideol + party * lin.score1
Model 2:count ~ (gender + party + ideol)^2
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1         7     8.3986
2         4     3.2454  3   5.1532   0.1609
```

The LR test statistic is 5.15 which gives a p-value of 0.16 on χ^2_3 (figure out the DF!). Thus, there is not sufficient evidence to conclude that the association is not linear. Practically speaking, the linear PI association model with these scores provides a reasonable fit to the data. (Note that this does not mean that the model or choice of scores is *correct*, as we could surely find other scores that are also not rejected.)

Notes

- The confidence intervals in this example demonstrate the other big advantage of making explicit use of ordinality:
  - *Confidence intervals for comparable odds ratios are generally much shorter in the ordinal analysis than in the nominal analysis*

  * For example the confidence interval comparing the two most extreme ideologies:
    - $(2.3 < OR < 10.3)$ in the nominal analysis
    - $(3.5 < OR < 9.0)$ in the ordinal analysis
- The ability to represent the PI association using fewer parameters allows those parameters to be estimated more precisely.
  * This precision is transferred to inferences about the odds ratios.

- It is not too difficult to apply ordinal association to the GI association instead of, or in addition to, the PI association
  - Left as exercise
- One can try using different scores if there is uncertainty about them
  - See below for advice on choosing scores.

4.2 Two ordinal variables

- When both variables in a two-variable model are ordinal we can define separate scores for each variable
  - $s_i^X, i = 1, \ldots, I$ for X and $s_j^Z, j = 1, \ldots, J$ for Z.
- In the regression model, the entire association is modeled by $\beta^{XZ} s^X s^Z$.
  - In abbreviated notation,

$$\log(\mu_{ij}) = \beta_0 + \beta_i^X + \beta_j^Z + \beta^{XZ} s_i^X s_j^Z$$

  - Continue to use nominal parameters for all variables' main effects to allow general pattern of means to be flexible
  - Easy to see that for this model $\log(OR_{ii',jj'}) = \beta^{XZ}(s_i^X - s_{i'}^X)(s_j^Z - s_{j'}^Z)$
    * This structure is sometimes called the *linear-by-linear association.*
    * The odds ratio depends on the two levels of X and the two levels of Z only through their respective differences in scores.
    * This is just like the interaction effect from two variables in a numeric Poisson regression
  - 1 parameter for entire association!
- The reduction of association parameters comes at the cost of more heavily structured assumptions regarding the nature of the log-odds ratios.
  - When a test for $H_0 : \beta^{XZ} = 0$ is rejected, it does not imply that the association between X and Z is *entirely* linear

- Similarly, failing to reject $H_0 : \beta^{XZ} = 0$ means only that there is insufficient evidence of a *linear-by-linear* association
  * There may be association that is not captured by the linear-by linear model
  * This is the same as when we assume linearity in linear regression
- Should test fit of this model compared to either the linear association or the nominal association models

5 Conclusions: What to learn

1. Ordinal vs. nominal variables
2. Scores: what they mean and how to choose them
3. Models involving scores
   (a) Formulate
   (b) Fit
   (c) Manipulate into ORs
      i. Pay attention to **subscripts** on parameters in all estimates of means, and ratios. They control *everything*!
4. Comparing models

6 Exercises (due when announced)

Recall the example we have been studying in Lectures 15 and 21 on the relationship between number of chronic diseases and self-reported health levels. Both of these variables are ordinal. We will now try to maker use of that fact. We have previously seen that the ratios of means for higher to lower health levels seemed to change systematically (decreasing) as the number of chronic diseases increased. Let's see if we can model this. **IN ALL PARTS BELOW, REPORT ONLY THE OUTPUT THAT IS REQUESTED. DUMPING UNNEEDED OUTPUT ON THE MARKER WILL RESULT IN DEDUCTIONS.**

1. Refit the nominal homogeneous association model from Lecture 21 using the version of the data where we combined 5–8 chronic diseases into "5+" to avoid problems with empty cells. Much of the problem below is related to what you did in Lecture 21. Feel free to repeat or adapt code from there.

   (a) Run the LR tests on each effect on the model. Report the test statistic and degrees of freedom for the interaction term.

   (b) Using `emmeans`(), estimate the log-means for each cell in the table. Nothing to report here. Show the output.

   (c) Compute ORs with CIs using consecutive numbers of diseases and all pairwise comparisons of health levels using higher/lower directions (`interaction=list("consec","pairwise")`) does this). Using the "none" adjustment for multiple inferences, print out the results.

   (d) Separately for each pair of health levels, look at the CIs for the ORs from consecutive numbers of diseases. Do the CIs all overlap? In other words, is there some region that all the CIs cover? To determine this, among each set of ORs comparing two levels of diseases for a given pair of health levels, report the largest lower confidence limit and the smallest upper confidence limit.

   (e) What does this last result suggest about the possibility of replacing all ORs for consecutive numbers of chronic diseases with a single value within each pair of health levels?

2. Using consecutive integer scores (0-1-2-3-4-5) on the disease groups, fit the linear association model, leaving health levels as nominal.

   (a) Run the LR tests on the interaction term. Report the test statistic and degrees of freedom for the new linear association term, and draw conclusions about the linear association between health levels and number of chronic diseases.

   (b) Use `ref_grid()` to equate the score variable to chronic diseases categorical variable. Then use `emmeans`() to get the means for all cells of the table. Print the results.

   (c) Repeat the OR computations from the previous model. If you do this right, you should have identical ORs and CIs for each change of 1 disease within each pair of health levels. Print out the results and interpret what each of the three OR estimates measures.

(d) Compare the confidence interval widths to those from the two nominal ORs. Are they generally narrower than before?

(e) Now let's see if there is information about the diseases-health association that is not explained by the linear association. Run the LR test to check the fit of the linear association model compared to the nominal association model. Report ALL parts of the test: hypotheses, test statistic, p-value, and conclusions. Are we comfortable that the linear-association model has explained the diseases-health association well?

3. Notice that the health variable is *also* ordinal! Use 1-2-3 scores on it along with the scores on the number of chronic diseases from the previous problem. Fir the a linear-by-linear association model

   (a) Compare fit of this model against the fit of the model we created in Q2. So we are testing only whether the quality of fit is damaged by *adding* scores for outcome to the previous linear association model with age scores. Report the test statistic, p-value, and conclusion.

   (b) Calculate the ORs. Note that `ref_grid` can use something like `cov.reduce=list(X_scores~X, Z_scores~Z)` to simultaneously equate X and Z scores to their respective levels.

      i. Print out all of the ORs. Look at the pattern of estimates, observing which ones are the same and which ones are different. Explain this pattern in terms of score differences.

STATISTICS 475: Applied Discrete Data Analysis

Lecture 24: Model and Variable Selection

(B&L Section 5.1)

1 Problem to be solved

- We have learned to fit and analyze regression models for binary and count responses
- We always start by assuming that we know which variables belong in the model
  - VERY often, we have many variables but don't know which ones are needed
  - Goal of research may be to identify important variables
- We need methods for deciding which variables are important
- In the process, we will also develop methods for comparing models that are not nested
- All of our models are GLMs that link the mean to the linear predictor in the same basic way,
$$g(\theta) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_p x_p$$
  - The process of determining variables operates the same way in all GLMs.

2 Model Comparison Criteria

- Suppose that the full model has a linear predictor (LP) $\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_p x_p$
- We have compared NESTED MODELS using LR tests
  - "Nested" means that the reduced model is a version of the full model with some constraints added on parameters
    - For example, $H_0 : \beta_1 = 0$ implies $H_0 : LP = \beta_0 + \beta_2 x_2 + \ldots + \beta_p x_p$
    - $H_0 : \beta_1 = \beta_2$ implies $H_0 : LP = \beta_0 + \beta_{(12)}(x_1 + x_2) + \ldots + \beta_p x_p$

    * Both of these LPs are nested within the full model
  - We cannot use this to handle non-nested model
    * $LP_1 = \beta_0 + \beta_1 x_1$ vs. $LP_2 = \beta_0 + \beta_2 x_2 + \beta_3 x_3$
    * Neither LP can be created by manipulating parameters on the other model
    * Each has something that is not contained in the other model
  - We presently have no way of comparing such models

- LR test is based on computing RESIDUAL DEVIANCE for each model
  - Residual deviance is $-2*$log likelihood evaluated at MLE
  - Measures how closely the estimated model "fits" the data
    * Like sum of squared errors, SSE, in linear regression
    * Residual deviance gets smaller if variables are added to the model, even if they not important
      · It just makes model move closer to noise in data ("overfitting")
      · Can be reduced to 0 if model is large enough ("saturated")
    * Therefore, residual deviance has limited use for comparing models of different sizes.

- INFORMATION CRITERIA (ICs) are "penalized" versions of residual deviance that can actually get larger if useless variables are added to model.
  - Suppose a model has r parameters (e.g., in a typical LP for p variables, $r = p+1$), and that its residual deviance is D
  - General form for an information criterion is

$$IC(k) = D + kr,$$

    where k is a "penalty" coefficient that we must choose
    * The smaller $IC(k)$ is, the "better" a model fits the data without overfitting
    * When you add variables to a model, D decreases, but r increases, so the penalty term kr determines how much smaller the deviance must become to believe that the variable is explaining more than just noise in the data
    * Using a larger k will tend to favour smaller models, because the penalty kr is larger

- The three most common information criteria are:
  1. Akaike's Information Criterion (AIC):

$$AIC = IC(2) = D + 2r$$

  2. Corrected AIC (AIC$_c$):

$$AIC_c = IC(2n/(n-r-1)) = D + \frac{2n}{n-r-1}r = AIC + \frac{2r(r+1)}{n-r-1}$$

3. Bayesian Information Criterion (BIC):

$$BIC = IC(\log(n)) = D + \log(n)r$$

   - Each one is useful; no one is best for all problems
     * Use BIC if you want to keep models small and explainable, and are willing to exclude variables that might make real, but very small improvements to the model
     * Use AIC is you want to have higher chance of including ALL useful variables, at the risk of also including some that are not useful
     * Use AIC$_c$ as a compromise: more likely to include most important variables, less likely to include many unimportant variables

2.1 Using Information Criteria

- Once we select a particular $IC(k)$ we need to evaluate it on some models to compare them
  - Which models?
- Often we have a specific set of models
  - Recall loglinear models, where we had heterogeneous association, homogeneous association, mutual independence, and models in between that included some but not all 2-way interactions
  - In other problems, past research might suggest specific models to compare
  - Then compute your $IC(k)$ on each model, report them all, and favour the model with the smallest value.
    * Models within 2 IC-units of the best are also competitive with the best
    * Even those within about 5 IC units are still somewhat supported by the data

Example: Political Ideology (Lecture 24 IC scripts.R, PolIdeolData.csv)

We return to the political ideology data, where we previously created models representing heterogeneous association (saturated), homogeneous association, mutual independence, and an ordinal association model that used 1-2-3-4-5 scores in a linear PI association. Now we refit those models and add two ordinal models, one using 0-2-3-4-6 to spread out the extremes, and one using 2-1-0-1-2 to measure an ordinal association with extremeness. We will compare these models using AIC, BIC, and AIC$_c$. Note that AIC$_c$ won't work on models with $n-1$ or n parameters, because the penalty term becomes infinite or negative in this case.

Several R functions compute ICs. We will use `extractAIC(<model>,k=)`, because it produces r and $IC(k)$ for any specified value of k as the in the `k` argument value, with `k=2` as the default. The model fits are all in the program. Below are the BIC computations. Computations for AIC and AICc are also shown. I first save n, computed as the number of rows in our data frame, then use its log as our penalty. Then I combine all of the ICs into a nicer-looking array for easy viewing.

```
> n.obs = nrow(alldata)
> (bic.sat = extractAIC(mod.sat, k=log(n.obs)))
[1]  20.0000 168.3451
> bic.hom = extractAIC(mod.homo, k=log(n.obs))
> bic.ind = extractAIC(mod.indep, k=log(n.obs))
> bic.lin1 = extractAIC(mod.homo.lin1.PI, k=log(n.obs))
> bic.lin2 = extractAIC(mod.homo.lin2.PI, k=log(n.obs))
> bic.extr = extractAIC(mod.homo.extr.PI, k=log(n.obs))
>
> allBIC = rbind(bic.sat, bic.hom, bic.ind, bic.lin1, bic.lin2, bic.e
> colnames(allBIC) = c("Params", "BIC")
> rownames(allBIC) = c("Sat", "Homo", "Ind", "Lin1", "Lin2", "Extr")
> allBIC
     Params      BIC
Sat      20 168.3451
Homo     16 159.6076
Ind       7 209.2520
Lin1     13 155.7736
Lin2     13 160.3287
Extr     13 210.9731
```

Remembering that smaller values indicate better-fitting models, we see that the `Lin1` model, which has homogeneous associations and a linear association for PI using 1-2-3-4-5 scores, has the smallest BIC ($BIC = 155.8$). It would be the preferred model from this group. Note that no other models are within 2 units of 155.8, so no other models have much support from the data, although two—`Homo` and `Lin2`—lie within 5 units and, therefore, are not completely discounted.

3 Variable selection

- In some cases, we have a pool of p variables and nothing to guide us in forming models
  - Literally any combination of variables might be "best," but we have no idea which one.
  - Notice that there are 2^p possible models we could form
    - $p = 10$: about 1 thousand models
    - $p = 20$: about 1 million models
    - $p = 30$: about 1 billion models
  - So this is only viable for small-to-moderate numbers of variables.

- We could try fitting them all and then choosing the one with the lowest $IC(k)$ for some chosen k
  - This is done surprisingly often! It's called ALL-SUBSETS REGRESSION
- All subsets regression is popular but flawed
  - Randomness in the data makes some models look better than they should and others look worse.
  - The problem grows as p grows
    * There is one model that is *actually* the best (in an infinite sample), and $2^p - 1$ that are not
  - Usually lesser models "get lucky" in explaining a particular sample of responses and outperform the actual best model
    * We shouldn't put too much faith in the results; it's just a suggestion
  - Estimating parameters and computing tests and confidence intervals using selected models often leads to biased results
    * Useless variables get selected when they have a lucky alignment with the response.
    * Parameter estimates and CIs are farther from 0 than they would be in other samples, look falsely important.
    * Variables left out of the model have parameter estimates of 0, with 0 standard error!
      · As if we KNEW that they didn't belong in the model, but then why did we include them in the variable selection
  - Reporting other models with similar $IC(k)$ values helps to express appropriate uncertainty about results

3.1 Bayesian Model Averaging

- Suppose that a "magic 8-ball" told you that it was 50% sure that a parameter value was 1.0, 30% sure it was 0.5, and 20% sure it was 1.5.
  - Rather than selecting one of these estimates, which has at best a 50-50 chance of being right, we could "hedge our bets" by taking an average, weighted by the chance that the answer is right
    * Take $50\% * 1.0 + 30\% * 0.5 + 20\% * 1.5 = 0.95$
    * This uses all of the information we have and turns out to be the theoretically optimal answer
- We can try doing the same thing with models
  - Use the $IC(k)$ values to rank or weight the fit of each model

- Create an averaged model that combines the linear predictors of all of the individual models, according to their weight
  - Models with large weight contribute strongly
  - Models with tiny weight hardly count at all.

- Bayesian Model Averaging (BMA) does exactly this
  - Each model's BIC is transformed into an estimated "posterior probability" that the model is truly the best one[1]
  - Then each parameter is separately averaged using these posterior probabilities as weights
  - Suppose that there are M models being considered (e.g., $= 2^p$ from all subsets regression), where for $m = 1, \ldots, M$, model m has
    - Parameter estimates $\hat{\beta}_{0,m} + \hat{\beta}_{1,m}x_1 + \hat{\beta}_{2,m}x_2 + \ldots + \hat{\beta}_{p,m}x_p$ where some of these are fixed at 0 if the variable is not included in model m
      - Estimates of a given parameter generally change depending on what other variables are in the model.
    - BIC value BIC_m
    - Posterior probability from BIC, $\hat{\tau}_m$
      - These add to 1 across all M models.
    - For each variable x_j, $j = 1, \ldots, p$, the BMA parameter estimate is

$$\hat{\beta}_j^{BMA} = \sum_{m=1}^{M} \hat{\tau}_m \hat{\beta}_{j,m}$$

  - A standard error can also be calculated using both the variability of parameter estimates across models and the estimated variances within each model
    - Confidence intervals can be found for each parameter that do not suffer from the biased interpretations that happen when only a single model is selected.

- Also notice: each variable shows up in exactly half of the models
  - There are 2^{p-1} models with x_j and 2^{p-1} models without x_j that are formed from the remaining $p - 1$ variables.
  - Each half of the models has a certain amount of total posterior probability, added across those models
  - The total probability from all models *with* x_j is the estimated posterior probability that x_j belongs in the best model (i.e., that $\beta_j \neq 0$)
    - High probabilities indicate important variables

[1] A posterior probability is a construct from Bayesian statistics that represents the probability of an event, given the data. In Bayesian statistics, we start with a "prior" belief about a problem (e.g., all models have equal probability of being best). We then use the data and some calculations to update this belief, which is called a "posterior". See STAT 460 to learn more about Bayesian statistics.

- Above 0.5: more likely than not to be important
- Above 0.75: better than 3:1 odds that variable is important
- Above 0.90: pretty certainly important (10:1 odds)

Example: Placekicking (Lecture 24 BMA scripts.R, Placekick.csv)

We return to our old friend, the placekick data. Recall the data:

```
> head(placekick)
  week distance change   elap30 PAT type field wind good
1    1       21      1 24.7167   0    1     1    0    1
2    1       21      0 15.8500   0    1     1    0    1
3    1       20      0  0.4500   1    1     1    0    1
4    1       28      0 13.5500   0    1     1    0    1
5    1       20      0 21.8667   1    0     0    0    1
6    1       25      0 17.6833   0    0     0    0    1
```

The variable `good` is a binary response, and other variables are explanatory. We now ask the question: Which variables appear to be important for the probability of a successful placekick? We will use BMA to answer this question. We will fit all possible combinations of the available variables (all subsets), compute the BIC for each, and estimate the posterior probability that each one is the best model.

The good news is that we don't have to do all of this manually. There are several R packages that do all-subsets and BMA. We will use the `MuMIn` (**Mu**lti **M**odel **In**ference), package for this. The process is as follows:

1. Fit the full model including all variables using `glm()`
   (a) MUST add `na.action=na.fail` to `glm()` or the rest won't work
2. Use the `dredge()` function to evaluate the chosen $IC(k)$ on each possible model. This produces a data frame with each model's parameter estimates, the BIC, and related statistics
   (a) Arguments: `global.model` is the model from step (1); `rank` is the $IC(k)$ (we use `"BIC"` here)
   (b) One useful statistic, `delta`, is the difference between a model's $IC(k)$ and the smallest $IC(k)$ among all models (helps with the <2, <5 guidelines)
3. Look at some of the models, either by printing out some rows of the data frame or by using the `subset()` function, where a second argument defines which roes of the matrix to select
   (a) We show `delta <= 5`
4. Use the `model.avg()` function to compute model-averaged parameter estimates and posterior probabilities for each variable

5. Manipulate the results and present them in a table

The analysis results are below:

```
> mod.fit <- glm(good ~ ., family = binomial(link = "logit"),
                           data=placekick, na.action = na.fail)
>
> allmods <- dredge(global.model=mod.fit, rank = "BIC")
Fixed term is "(Intercept)"
> allmods[1:10,]
Global model call: glm(formula = good ~ ., family = binomial(link = "
                       data = placekick, na.action = na.fail)
---
Model selection table
    (Intrc)   chang    dstnc    elp30     field   PAT
19    4.516         -0.08587                     1.338
20    4.676 -0.3402 -0.08646                     1.245
147   4.599         -0.08676                     1.319
83    4.767         -0.08592                     1.339
3     5.812         -0.11500
51    4.445         -0.08550                     1.343
23    4.464         -0.08548 0.003274            1.340
27    4.521         -0.08589          -0.009337 1.338
4     5.893 -0.4478 -0.11290
148   4.752 -0.3351 -0.08724                     1.230
      type     week    wind df   logLik   BIC delta weight
19                          3 -381.205 784.2  0.00  0.687
20                          4 -379.666 788.4  4.18  0.085
147                 -0.5326  4 -379.831 788.7  4.51  0.072
83          -0.02597         4 -380.285 789.6  5.42  0.046
3                           2 -387.873 790.3  6.07  0.033
51  0.0805                  4 -381.126 791.3  7.10  0.020
23                          4 -381.156 791.4  7.16  0.019
27                          4 -381.204 791.5  7.26  0.018
4                           3 -385.250 792.3  8.09  0.012
148                 -0.5234  5 -378.343 793.0  8.80  0.008
Models ranked by BIC(x)
> subset(allmods, delta <= 5)
Global model call: glm(formula = good ~ ., family = binomial(link = "
                       data = placekick, na.action = na.fail)
---
Model selection table
    (Intrc)   chang    dstnc   PAT    wind df   logLik
19    4.516         -0.08587 1.338          3 -381.205
20    4.676 -0.3402 -0.08646 1.245          4 -379.666
```

```
147   4.599          -0.08676 1.319 -0.5326  4 -379.831
       BIC delta weight
19  784.2  0.00  0.814
20  788.4  4.18  0.100
147 788.7  4.51  0.085
Models ranked by BIC(x)
```

In this part, we show the original fit in `glm()` and the results produced by `dredge()`. The matrix is sorted by BIC, from smallest top largest, which means the posterior probabilities are largest to smallest. The top 10 models are printed first, 1 per row (wrapped into a second set of lines), where a blank under a variable name means that the variable was not in that model. We can see that *all* of the best models include `distance`, most include `PAT`, and the rest of the variables are not consistently included in the models. The best model has $BIC = 784.2$, and the posterior probability estimate, under `weight`, is 0.69. That mean that, after looking at the data, we believe that there is a 69% chance that `distance+PAT` is the best model. The parameter estimates for `distance` vary from about -0.085 to -0.115 across these 10 models, where the value changes more when `PAT` is left out of the model than anything else. We see in the second table that there are two additional models with `delta` values below 5, but none below 2. This suggests that there is only a little bit of competition from any other specific model for the best one.

Next we look at the model-averaging results.

```
> # Model averaging with model.avg
> MA <- model.avg(object = allmods)
> # Models and posterior probabilities ("weight") are in msTable
> MA$msTable[1:10,]
     df    logLik      BIC    delta     weight
25    3 -381.2049 784.1956 0.000000 0.65688254
125   4 -379.6662 788.3802 4.184578 0.08106215
258   4 -379.8309 788.7094 4.513828 0.06875786
257   4 -380.2855 789.6186 5.423004 0.04364126
2     2 -387.8725 790.2689 6.073279 0.03152768
256   4 -381.1263 791.3004 7.104740 0.01882405
235   4 -381.1557 791.3591 7.163511 0.01827894
245   4 -381.2037 791.4550 7.259429 0.01742299
12    3 -385.2498 792.2853 8.089662 0.01150376
1258  5 -378.3433 792.9961 8.800516 0.00806269
> # Variable weights
> weights <- MA$sw
> # Model-averaged coefficients in table
> round(summary(MA)$coefmat.full, digits = 3)
            Estimate Std. Error Adjusted SE z value
(Intercept)    4.626      0.551       0.551   8.390
distance      -0.088      0.013       0.013   6.882
```

```
PAT              1.252      0.480        0.481   2.605
change          -0.042      0.133        0.133   0.317
wind            -0.051      0.184        0.184   0.276
week            -0.002      0.008        0.008   0.204
type             0.003      0.038        0.038   0.068
elap30           0.000      0.002        0.002   0.052
field            0.000      0.031        0.031   0.012
            Pr(>|z|)
(Intercept)    0.000
distance       0.000
PAT            0.009
change         0.752
wind           0.783
week           0.838
type           0.946
elap30         0.958
field          0.990
> # Confidence intervals for parameters
> round(cbind(coef(MA), confint(MA)), digits = 3)
                     2.5 % 97.5 %
(Intercept)  4.626  3.546  5.707
distance    -0.088 -0.113 -0.063
PAT          1.326  0.578  2.074
change      -0.354 -0.740  0.031
wind        -0.535 -1.150  0.081
week        -0.026 -0.064  0.012
type         0.091 -0.315  0.497
elap30       0.003 -0.017  0.024
field       -0.014 -0.388  0.359
> # Exponentiating into ORs, removing intercept
> round(cbind(exp(coef(MA)[-1]),
+             exp(confint(MA)[-1,]),
+             weights), digits = 2)
              2.5 % 97.5 % weights
distance 0.92  0.89   0.94    1.00
PAT      3.77  1.78   7.96    0.94
change   0.70  0.48   1.03    0.12
wind     0.59  0.32   1.08    0.09
week     0.97  0.94   1.01    0.06
type     1.10  0.73   1.64    0.03
elap30   1.00  0.98   1.02    0.03
field    0.99  0.68   1.43    0.03
```

The table produces by `model.avg()` gives the same model statistics as in the matrix from `dredge()`, but now the model numbers contain the variable numbers in alphabetical order (a bit annoying, but here, `distance` is second after `change` and `PAT` is fifth). The `sw` component of the saved object contains the posterior probabilities for each of the variables, which we show in the last column of the last table. We can run the `coef()` and `confint()` functions on the object to get model-averaged parameter estimates and confidence intervals. These are all on the parameters in the linear predictor scale (change in log-odds for 1-init increase in x_j), so we exponentiate them into odds ratios.

The last table slows the odds ratio estimates and their respective confidence intervals, along with the posterior probabilities for each variable. First, we see that only `distance` and `PAT` have large posterior probabilities (rounded to 1.0 and 0.94, respectively), with a HUGE jump down to the next best variable, `change` (0.12). So it is fairly clear which variables matter here. Also, the confidence intervals for odds ratios reinforce this, with only these two excluding 1. A 1-yard increase in distance is associated with an estimated OR of 0.92, or about an 8% reduction in odds of success. When a kick is a PAT, the estimated odds of success are 3.8 times as high as when it is a regular field goal.

4 Conclusions: What to learn

- All models are wrong, but some are better than others
- Information criteria are penalized versions of residual deviance that account for the size of the model (number of parameters) to allow fair comparisons between models of different sizes, even if the models are not nested.
- All subsets regression is possible but has problems
- Bayesian Model Averaging produces parameter estimates and confidence intervals that make better sense than ones from a single selected model
  - Also get posterior probabilities on models and individual variables

5 Exercises (due when announced)

1. Refer to the Abalone data. Perform all-subsets regression using a Poisson GLM and log link with all variables in their original, linear form. Since Sex is a 3-level categorical variable entered as numeric, replace it in your data frame with two indicators:

```
aba$sexf = ifelse(aba$Sex==2, yes=1, no=0)
aba$sexm = ifelse(aba$Sex==1, yes=1, no=0)
```

   (a) Identify the top model. Which variables are in it and what is its BIC?
   (b) What are the BIC values for the top 5 models?
   (c) How many models are with 2 and 5 BIC units of the best?

2. Repeat Exercise 1 using AIC (literally just change the criterion to `AIC` in the `rank` argument of `dredge()`

3. Use BMA to compute posterior probabilities for each model starting from the all-subsets with BIC.
   (a) Present the top 5 models and look at their posterior probabilities. Is it fairly obvious that there is one best model, or is there a fair amount of uncertainty?
   (b) Compute model-averaged parameter estimates for each variable.
      i. Show the posterior probabilities for the variables and interpret the results
      ii. Show a table of the parameter estimates and confidence intervals
      iii. Averaged across all models, by how much does the mean number of rings change for each 1-unit increase in shell weight, and what is the 95% confidence interval for this parameter?

STATISTICS 475: Applied Discrete Data Analysis

Lecture 25a: Residuals and Goodness-of-Fit Statistics (Abridged)

(B&L Section 5.2)

1 Problem to be solved

- All models are wrong. Some are useful.
  - We want to use models that fit the data well enough to be useful
    - True probabilities may not exactly follow a sigmoidal logistic curve, but are they close?
    - True log-mean counts may not be linear in explanatory variables, but are they close?
    - Counts may not be exactly Poisson or Binomial, but are they close?
- We can look at plots, but they are of limited use
  - Plotting probabilities or means only works in 1-variable models...what to do in other cases?
- We can look at quantitative measures of fit, but they, too are limited
  - Model fit may have many aspects; hard to capture in a single number.
- Kind of have to do a little bit of everything here

2 Review of Residuals in Linear Regression (Tom: mostly SKIP; Students: Read as necessary)

- **In linear regression, we *assume* the model,**

$$Y = \beta_0 + \beta_1 x_1 + \ldots + \beta_p x_p + \epsilon,\ \ \epsilon \sim N(0, \sigma^2)$$

- In other words, we assume that
  - The mean changes linearly with each explanatory variable
  - The errors are normally distributed
  - The variances are constant
  - ...and some other things
- The model fitting process does not tell us by itself whether the model we have fit is a good one.
- **We used residuals, $e_i = y_i - \hat{y}_i$, to identify model assumptions that may be inappropriate for the data**
  - **We plotted the residuals vs. $\hat{y}$ to see whether there is curvature or changing variance or maybe one or two outliers**
  - **We plotted residuals vs. each x_j to see the same thing**
- Standardized versions, $r_i = e_i/\sqrt{\widehat{Var}(e_i)}$, are also used, to make it easier to spot problems
  - Standardized residuals should behave (very) approximately like a sample from a standard normal distribution
  - Should have only about 5% of residuals in sample beyond ± 2, and about 1/1000 beyond ± 3
- We can use residuals similarly for all of our count models
  - Need to define residuals in context of each model
  - Need to identify assumptions to be checked

3 Residuals for count-data models

We will work only with Poisson and binomial models here

3.1 Aggregate the data

- Count data are discrete: they take on only integer values
  - As few as two possible values!
  - Implies that residuals might have only two possible values at each x—not very informative
- Start by aggregating all responses and predicted values as *counts*, with one total count for each different combination of explanatory variables *that will be used in the largest model being fitted*

- Aggregate data into EXPLANATORY VARIABLE PATTERN (EVP) FORM
  - Among n observations, there are M unique values (or combinations of values) of explanatory variables
    - In placekick data, n =1425 binary trials, but only M =43 different distances (where we once made individual confidence intervals)
    - Each unique combination is called an EXPLANATORY VARIABLE PATTERN (EVP)
  - Aggregate counts for each different combination of explanatory variables into a total as we have done before.
    - Call the aggregated count w_m, $m = 1, \ldots, M$
  - Record how many of the original data counts or trials made up each total (call this n_m to mirror w_m)
    - If every observation has a unique EVP, then $M = n$, $n_m = 1$ for all m, and $w_m = y_i$

- Examples of Aggregation to EVP form **(Tom Skip)**
  - In logistic regression, original data may consist of either binary responses or aggregated counts of successes and failures or counts of successes and trials
    - e.g., in Placekicking example with distance as only variable, have one count of trials (n_m) and successes (w_m) at each distance
  - In Poisson regression, aggregate counts if there are multiple observations for any combination of explanatory variables (for any "EVP")
    - For, example, in the left side of the table below, there are repeated counts for some combinations of x_1 and x_2.
    - If we are planning to use both variables in a model, then we need to aggregate according to distinct combinations of both variables (an EVP is a distinct combination of x_1 and x_2)
    - These are combined into a single total in the table on the right, and the number of counts that were added together is recorded as n_m, $m = 1, \ldots, 6$

$y_i\ i = 1, \ldots, n$	x_1	x_2	$\longrightarrow$	w_m, $m = 1, \ldots, 6 = M$	x_1	x_2	n_m
2	1	1		2+7=9	1	1	2
4	1	2		4	1	2	1
7	1	1	$\longrightarrow$	5+4=9	2	1	2
5	2	1		3	2	2	1
3	2	2		9	3	1	1
4	2	1		6+4=10	3	2	2
6	3	2					
4	3	2					
9	3	1					

- Once we have the aggregated data, we fit the model

- Use `glm()` with `formula=w/n~...` and `weights=n` to account for the number of trials or counts that went into the total
  - Yes, now adding `weights=` to Poisson.
  - Models the *mean response* (probability π_m or mean count μ_m) at each EVP
- For each EVP get one predicted count $\hat{w}_m$ from the model
  - Poisson: $\hat{w}_m = n_m \hat{\mu}_m$
  - Binomial: $\hat{w}_m = n_m \hat{\pi}_m$
- So we now have $w_m, \hat{w}_m$ for $m = 1, \ldots, M$, where M is the number of EVPs

3.2 Compute residuals

- In each model, the variance of W depends on the mean of W
  - Binomial: $Var(W) = n\pi(1-\pi)$, where π is the probability of success for a single trial
  - Poisson: $Var(W) = n\mu$, where μ is the mean count for a single observation prior to aggregation
- Therefore the "raw" residual, $w_m - \hat{w}_m$ has variance that is not supposed to be constant
  - Different from linear regression
  - Makes interpreting "large" and "small" residuals difficult
- Standardized Pearson Residuals

$$r_m = \frac{w_m - \hat{w}_m}{\sqrt{\widehat{Var}(W_m - \hat{W}_m)}},\ m = 1, \ldots, M$$

where $\widehat{Var}(W_m - \hat{W}_m)$ is the estimated variance of the residual $W_m - \hat{W}_m$ based on the model

  - These are exactly the same quantities that we used for helping to understand association in a contingency table
    - In that context they measured how far an observed table cell count differed from the "expected count"
    - The "expected count" assumed independence, which we now understand is equivalent to loglinear mutual independence model
  - Now just using the same approach to measure how far the modeled counts are from the observed counts in more general models.
  - Same interpretation:
    - $\approx$5% of standardized residuals can be expected beyond ± 2,
    - rarely beyond ± 3,
    - never beyond ± 4

Example: Placekicking (Lecture 25 scripts.R, Placekick.csv)

We return to our old friend, the placekick data and will compute residuals for the logistic regression model using only `distance` as an explanatory variable. Recall the data:

```
> head(placekick)
  week distance change  elap30 PAT type field wind good
1    1       21      1 24.7167   0    1     1    0    1
2    1       21      0 15.8500   0    1     1    0    1
3    1       20      0  0.4500   1    1     1    0    1
4    1       28      0 13.5500   0    1     1    0    1
5    1       20      0 21.8667   1    0     0    0    1
6    1       25      0 17.6833   0    0     0    0    1
```

First, we need to aggregate the data into counts of successes and trials for each EVP. In this case, the model has just `distance`, so an EVP is just a different value of `distance`. This is the same w, n aggregation we have done before:

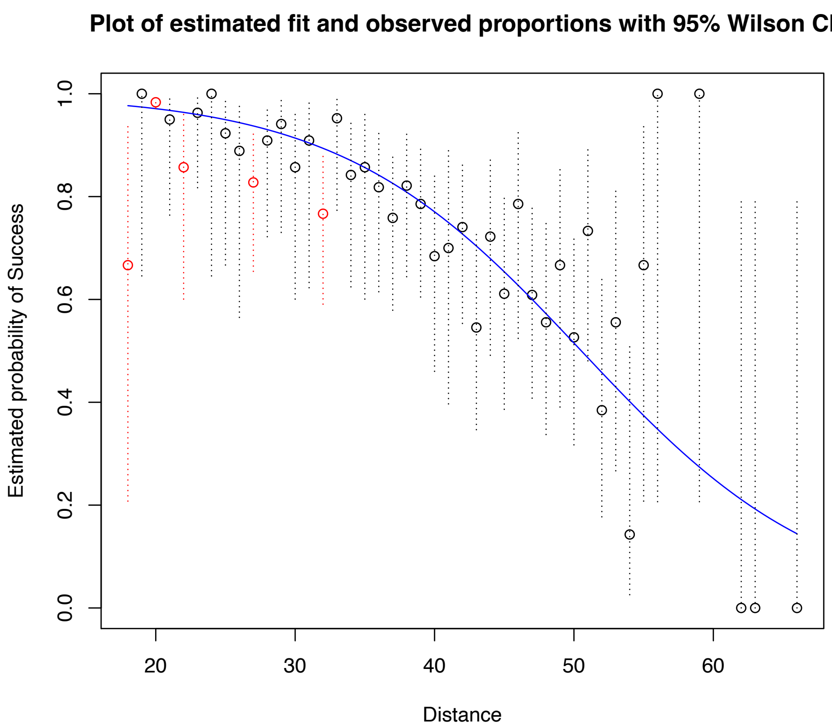
```
> w <- aggregate(x = good ~ distance, data = placekick,
+                  FUN = sum)
> n <- aggregate(x = good ~ distance, data = placekick,
+                  FUN = length)
> w.n <- data.frame(distance = w$distance, w = w$good,
+                     n = n$good, prop = round(w$good/n$good,4))
> head(w.n)
  distance success trials   prop
1       18       2      3 0.6667
2       19       7      7 1.0000
3       20     776    789 0.9835
4       21      19     20 0.9500
5       22      12     14 0.8571
6       23      26     27 0.9630
```

We then fit the model using `glm()`:

```
> mod.fit.bin <- glm(formula = w/n ~ distance,
+                      weights = n,
+                      family = binomial(link = logit), data = w.n)
```

The model fit is plotted in Figure 1, along with the Wilson CIs at each distance. I have added red colour to the CIs that do not cover the fitted curve, just to see better where these occur and how many there are. Five out of 43 miss the curve, which seems like more that I'd expect (5% of 43 is about 2). These are distances where we might expect that residuals will have large magnitudes.

Figure 1: Observed proportions and fitted model for placekick data using `distance` as only variable.



Standardized Pearson residuals for glm-class objects are available from `rstandard()`. We apply it to our model object below, and also get out the predicted values ($\hat{\pi}$'s and $\hat{w}_i$'s) and the estimated linear predictors. We add these into our data frame `w.n`. Then we print out the lines of data for any extreme looking residuals (absolute values beyond 2).

```
> pi.hat <- predict(mod.fit.bin, type = "response")
> w.hat <- w.n$n * pi.hat
> s.res <- rstandard(mod.fit.bin, type = "pearson")
> lin.pred <- mod.fit.bin$linear.predictors
> w.n <- data.frame(w.n, pi.hat, w.hat, lin.pred, s.res)
> round(head(w.n), digits = 3)
  distance   w   n  prop pi.hat   w.hat lin.pred  s.res
1       18   2   3 0.667  0.977   2.930    3.742 -3.575
2       19   7   7 1.000  0.974   6.819    3.627  0.433
3       20 776 789 0.984  0.971 766.130    3.512  3.628
4       21  19  20 0.950  0.968  19.352    3.397 -0.448
5       22  12  14 0.857  0.964  13.493    3.281 -2.149
6       23  26  27 0.963  0.960  25.908    3.166  0.091
> # Just the questionable residuals
> round(w.n[which(abs(w.n$s.res)>2),], digits=3)
   distance   w   n  prop pi.hat   w.hat lin.pred  s.res
1        18   2   3 0.667  0.977   2.930    3.742 -3.575
3        20 776 789 0.984  0.971 766.130    3.512  3.628
5        22  12  14 0.857  0.964  13.493    3.281 -2.149
10       27  24  29 0.828  0.937  27.185    2.706 -2.476
15       32  23  30 0.767  0.894  26.817    2.131 -2.299
```

We see a few potential problems among the standardized residuals: 2 values over 3.5 (which should be pretty rare—2/10,000 chance, totally unexpected when $M = 43$), another three more over 2. Notice that these are all at distances where the corresponding CI was not intersected by the fitted curve (I don't know if this is always the case...). One of these, 20 yards, is unusual because it contains almost all of the special PATs that are worth only one point. That EVP has its observed proportion of successes underestimated by the model (positive residual), while those at 18, 22, 27, and 32 are overestimated. So possibly the model is being drawn partway up toward the 20-yard EVP by the huge number of trials it represents, causing it to overestimate nearby EVPs that had mildly unusual low proportions of successes.

NOTE:

- The standard normal interpretation holds *well* only when the *model-estimated* counts of successes and failures are both at least 5:
  - $\hat{w}_m \geq 5$ and $n_m - \hat{w}_m \geq 5$

- All but one of the large residuals above fail this!!!
    - The first one fails it in both directions—be extra cautious about believing that one
    - Additional calculations shown in the book indicate that it is not *nearly* as unusual as the normal probability beyond -3.6 would suggest

3.3 Residual Plots

Residual Plots are used to identify problems in the residuals more easily than looking at a long list

r vs. $\hat{\mu}$ (Poisson) or r vs. $\hat{\pi}$ (binomial)

- If model fits well, plot should show a pretty random "cloud" of points
    - No serious fluctuations in the mean value
        - No obvious curvature
        - No long runs of mostly positive or mostly negative residuals
    - Roughly constant variance
    - $\approx$5% of points beyond ± 2, rarely any beyond ± 3, none beyond ± 4
        - IF the standard normal approximation is OK
- Strong curvature suggests the link (log or logit) specification needs to be changed
    - Or if there is only one explanatory variable, might need a transformation or polynomial terms in that variable.
    - **Plotting r vs. the *linear predictor*,** $\hat{\beta}_0 + \hat{\beta}_1 x_1 + \ldots + \hat{\beta}_p x_p$, can sometimes help diagnose how it needs to be changed.
- Changing variance (a funnel shape) indicates possible flaw in model for the probability distribution
    - In particular, a changing variance suggests that the data possess a different relationship between variance and mean than the model expects
        - May need to use a different distribution for model
- Points beyond thresholds scattered randomly across the plot
    - "One or two" past ± 3: outliers
        - By definition, outliers are very rare.
    - If there are much more than 5% beyond ± 2 and more than expected beyond other thresholds, something is more seriously wrong
        - Maybe a missing explanatory variable

* Maybe "overdispersion", especially if large values are scattered around the width of the plot.
  * OVERDISPERSION is another flaw in the distribution model where the actual variance in the data is not related to the mean the way the model thinks it should be

r vs. each x_j

- If there is only one explanatory variable, then general shape of plot is same as for plots vs. mean.
- Otherwise, looking mainly for curvature
  - curvature for a particular x_j plot suggests that a transformation or additional polynomial terms are needed *for that variable*
- If many plots show curvature, possibly the log or logit link specification needs to be changed.

Smoothing the mean trend in plots

- I like to add in a LOESS CURVE[1] to put a smooth weighted average of nearby points across the plot.
  - This makes it easier to identify possible curvature in the plot and also to see where there may be a trend of positive or negative residuals, indicating a poor fit to the link or to a particular x_j.

Interpreting the plots:

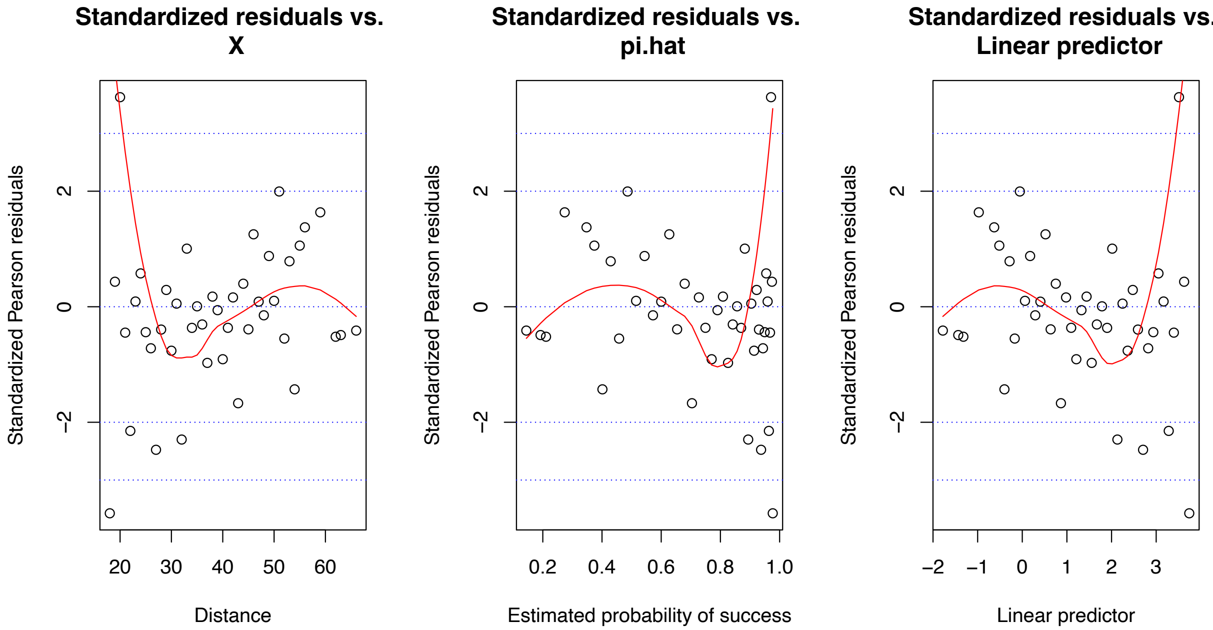
- If all is well with the model, I expect the LOESS curve to wobble randomly "not far" from 0
  - Should NOT get close to ± 2, and probably should stay well within ± 1, maybe? Maybe even less?
  - However, LOESS curves are unreliable near the edges of the plot, so shouldn't over-interpret trends on the first and last 10% of the data

Example: Placekicking (Lecture 24 scripts.R, Placekick.csv)

Refer to the plots in Figure 2. Note that we have added reference lines at $\pm 2, \pm 3$ so that it is easy to see where *potentially* extreme residuals are.

[1] LOESS (sometimes LOWESS) is short for "locally weighted scatterplot smoother. Think of it as sort of like a moving window that takes a vertical slice of the plot, computes the average Y value of the points within the slice, and plots the mean at the middle X value within the slice. Now slowly slide the window from left to right across the plot, continually computing the mean Y within the window and plotting it. This is one crude version of LOESS. The R function uses a better version that puts weights on points and computes a weighted average rather than completely including or excluding points in the mean computation.

Figure 2: Residual Plots for the Placekick Data



- All three plots shoot up toward the large residual at 20 yards that has so much weight.
- Otherwise, main central tendency actually seems to be a single slope, upward in plot vs. x, downward in plots against $\hat{\pi}$ and the linear predictor
  - This worries me: patterns like this suggest model flaws of some kind

Conclusion: There is something odd about that extreme positive residual.

- Contains over half of the placekicks
- Nearly all are PATs!!!
- Maybe there is something different about PATs vs. field goals??
- Try adding `PAT` to model to distinguish these kicks from regular field goals

---

4 Goodness-of-Fit Statistics

- Plots require interpretation
  - Subjectivity and experience influence interpretation
  - Numerical summaries would be nice.
- *Goodness-of-fit (GOF) statistics* are numerical summaries of the model fit

- Objective rather than subjective
- Summarize many aspects of fit into one number
- Provide no information regarding the *cause* of any poor fit that they might detect.
- Looking for so many things at once that they may not be good at finding any of them
  - Mainly indicate when there are *serious* problems

- Use in combination with residual plots
  - If GOF statistic indicates problems, plots tell you what they are
  - If plots "suggest" problems, GOF statistic helps gauge the severity
    - However, GOF statistic may not show problems if a model "mostly" fits well but has certain flaws seen in a plot

Residual Deviance

- Residual deviance, D, compares model-estimated counts to observed counts using log-likelihood
  - It is -2 times the difference in log likelihood between the fitted model and the saturated model
  - Smaller values = closer fit
- Residual degrees of freedom $df_r = M - (p+1)$
  - = Fitted points minus number of parameters estimated
- Deviance/DF statistic, D/df_r **is often computed**
  - If model is correct, expect $D/df_r \approx 1$, larger values indicate poor fit
  - If df_r is "not too small" we can suggest *very rough* thresholds for D/df_r based on the asymptotic $\chi^2_{df_R}$ approximation for D and the normal approximation to χ^2_ν when ν is large.
    - $D/df_r > 1 + 2\sqrt{2/df}_r$ indicates a *potential* problem
    - $D/df_r > 1 + 3\sqrt{2/df_r}$ indicates a *likely* poor fit
- The statistic tells you nothing about the cause of the problem
  - Use residual plots to figure out what is wrong
  - Can help you decide whether things you think you see in the plots might be serious
- *Deviance/DF must be computed on model fit to aggregated data*

<u>Example</u>: Placekicking (Lecture 25 scripts.R, Placekick.csv)

We compute the Deviance and Pearson statistics on the placekicking data using the model with `distance` only that we fit before. Below is part of the `summary()` output using a model fit to data that were previously aggregated by `distance`. Note the `Residual deviance` line in the results.

```
> summary(mod.fit.bin)

Call: glm(formula = success/trials ~ distance,
              family = binomial(link = logit),
              data = w.n, weights = trials)

<Output edited>

    Null deviance: 282.181  on 42  degrees of freedom
Residual deviance:  44.499  on 41  degrees of freedom
AIC: 148.46
```

We can extract the deviance and DF and also compute the thresholds automatically.

```
> rdev <- mod.fit.bin$deviance
> dfr <- mod.fit.bin$df.residual
> ddf <- rdev/dfr
> thresh2 <- 1 + 2*sqrt(2/dfr)
> thresh3 <- 1 + 3*sqrt(2/dfr)
> round(c(rdev, dfr, ddf, thresh2, thresh3), digits=2)
[1] 44.50 41.00  1.09  1.44  1.66
```

The residual deviance is 44.50 on 41 degrees of freedom. That leads to $D/df_r = 44.50/41 = 1.09$. The two thresholds for potential and likely poor fit are $1 + 2\sqrt{2/df}_r = 1.44$ and $1 + 3\sqrt{2/df}_r = 1.66$. Since 1.09 is well below both of these measures, the residual deviance suggests that there are no serious problems with this model. Notice that this is contrary to our analysis of the standardized Pearson residuals, which showed several concerns.

---

5 Conclusions: What to learn

- Residuals are one step in assessing model fit
- Important, because they provide detailed information about the model fit
- A little hard to interpret, because of subjectivity

  - Hard to judge how much curvature in LOESS is "too much to be from chance" without lots of practice

- Even objective measures of "Large" and "Small"—like the 2-3-4 thresholds—are not totally reliable all the time
  - Check exact probabilities if in doubt.
- Be able to make and interpret plots
- Be able to identify problems in the plots
- Deviance/df statistics can help a little for quantifying when problems might be present
  - They are not brilliant and have lots of limitations

6 Exercises (due when announced)

1. Refer to the Abalone data where we modeled Y = `Rings` against x = `Shell`. We started with a model that assumed that the log mean was linear in x.

   (a) Aggregate the data by shell weight. To show that you have done it right, report M and run `summary()` separately on w_m, $m = 1, \ldots, M$, and on n_m, $m = 1, \ldots, M$.

   (b) Refit the model to the aggregated data and show that it gives you the same results as in the example from Lecture 19, page 5.

   (c) Compute the standardized Pearson residuals and present a list of all values that are beyond ± 3, along with their respective values of w_m, n_m, x_m and $\bar{w} = w_m/n_m$, the mean count at each. How many are there? About how many would you expect there to be in a data set of size M?

   (d) Plot the standardized Pearson residuals against $\hat{\mu}$, and include a LOESS curve. Interpret this plot: does it seem to suggest that the model is a good fit? If not, what seems wrong?

   (e) Compute the Pearson/DF and Deviance/DF statistics. Interpret the results.

2. Repeat Exercise 1 for the model that used $\log(x)$.

3. Refer to the Salamander data in Exercise 2 of Lecture 19. Perform a complete diagnostic analysis (everything in this lecture) on the model of the number of salamanders against the number of years since the last burn. Be sure to draw conclusions supported by some evidence.